

Lab 04

Energy Operations Weather Agent

Microsoft Copilot Studio — Intermediate Lab Walkthrough

Duration: 2.5 hours | Difficulty: Intermediate

Skills Covered: Agent Creation, Topics & Variables, Connectors,
Power Fx, Custom Prompts, Connected Agents, Evaluations

Generated: June 2026

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Introduction

In this lab, you will build an Energy Operations Weather Agent that demonstrates how Microsoft Copilot Studio can integrate real-time external data sources with AI-powered analysis to deliver actionable business intelligence.

What you will build:

- A conversational agent that retrieves live weather data via MSN Weather connectors
- A custom AI prompt tool that generates energy grid operations briefings
- A connected child agent (Crew Safety Advisor) for specialized crew safety recommendations
- An evaluation test suite to validate agent quality at scale

Why this matters:

Energy utilities must make rapid operational decisions based on weather conditions. This agent demonstrates how AI can transform raw weather data into structured operations intelligence — reducing decision latency from hours to seconds.

Prerequisites

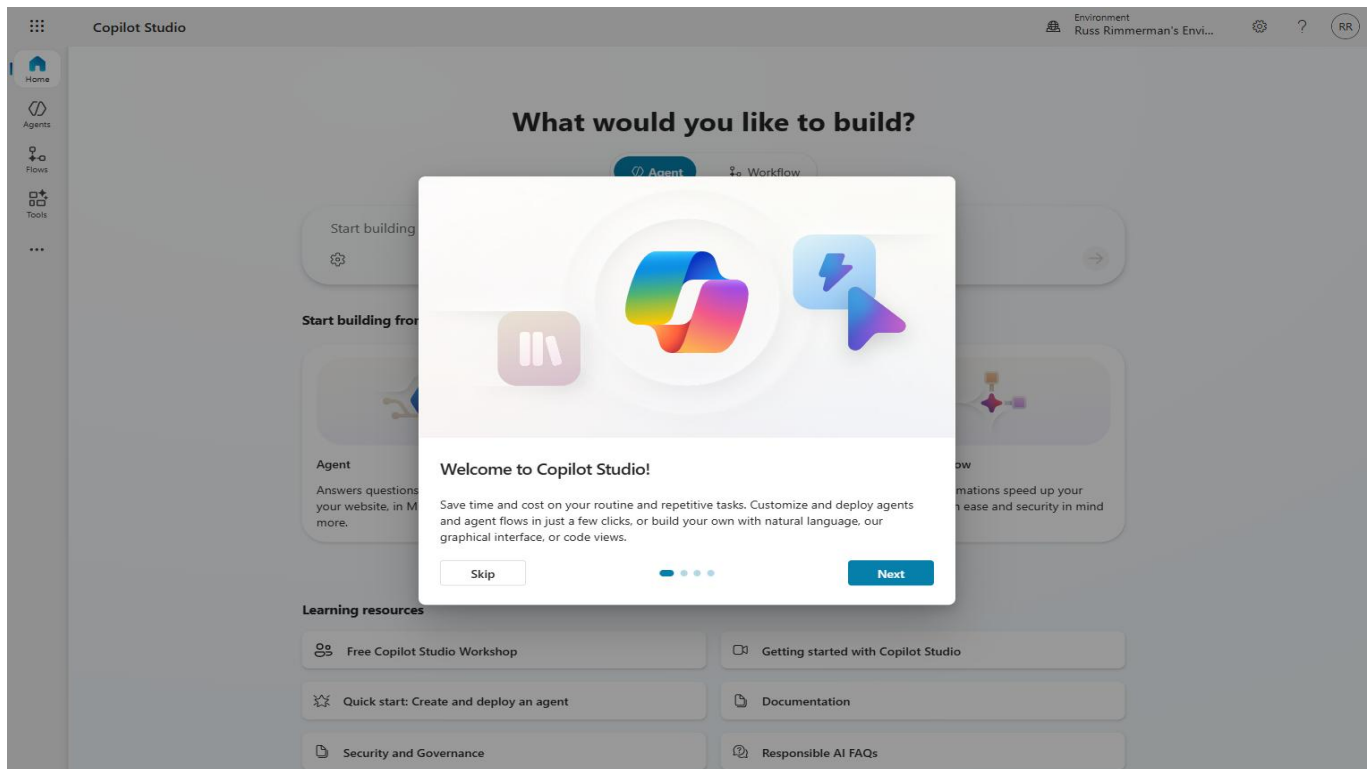
- Microsoft Copilot Studio access (copilotstudio.microsoft.com)
- Power Platform environment with Dataverse
- MSN Weather connector access (included with most M365 licenses)
- Basic familiarity with Copilot Studio interface (Lab 01-03 recommended)

Section 1: Create the Agent

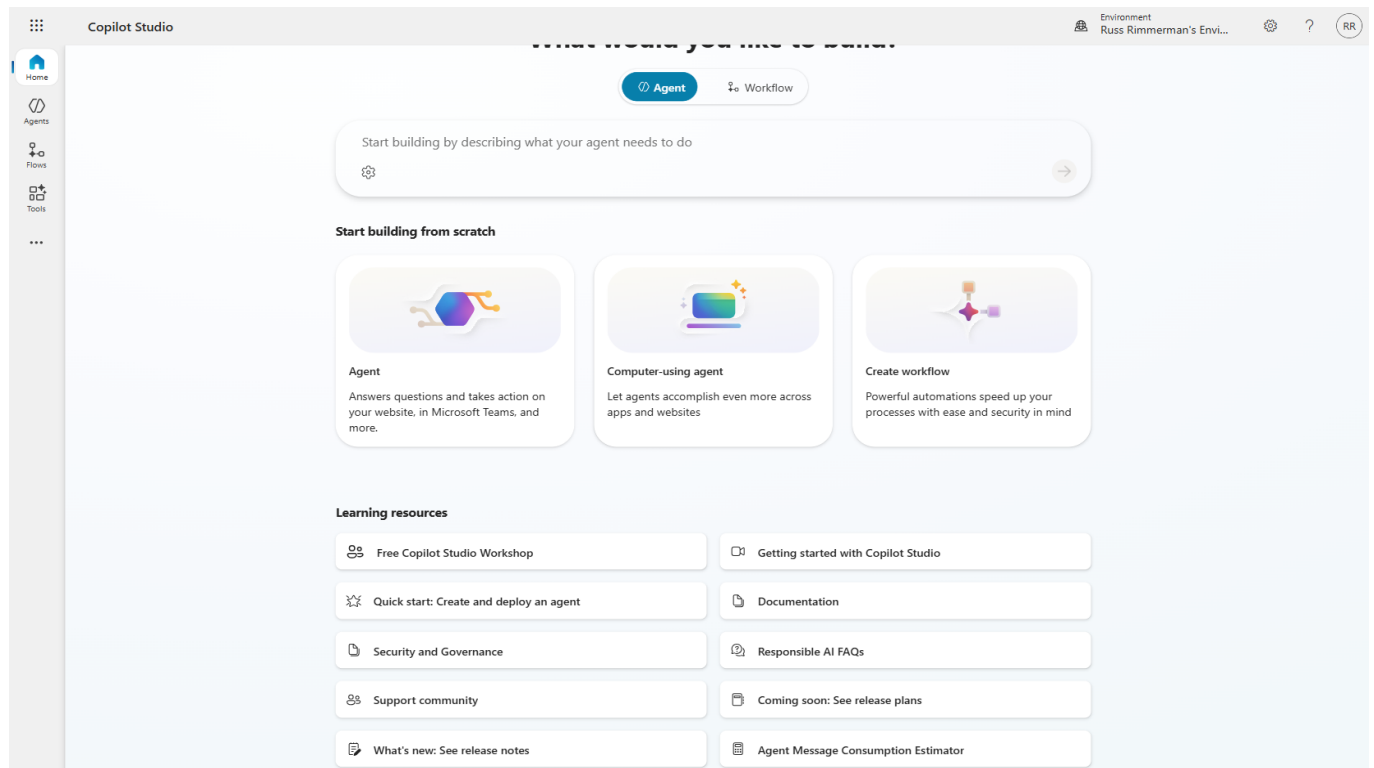
Every Copilot Studio solution starts with creating an agent. The agent is the top-level container that holds your topics, tools, knowledge sources, and configuration. In this section, you will create the Energy Operations Weather Agent with specialized instructions that guide the AI's behavior for energy grid scenarios.

1.1 Navigate to Copilot Studio

Step 1: Open your browser and navigate to copilotstudio.microsoft.com. Ensure you are signed into your Power Platform environment.

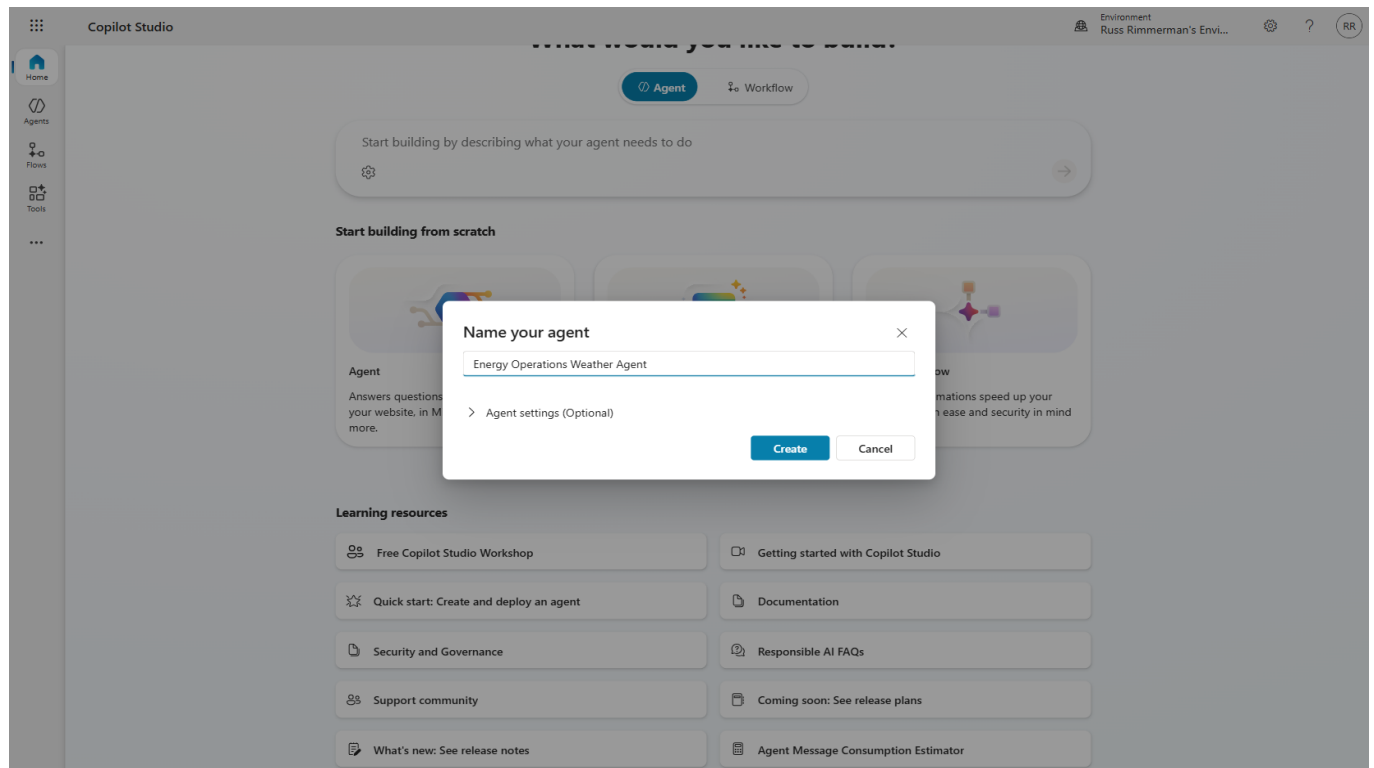


Step 2: From the Home page, click "New agent" or select "Agents" from the left navigation. You'll see the agent creation interface.



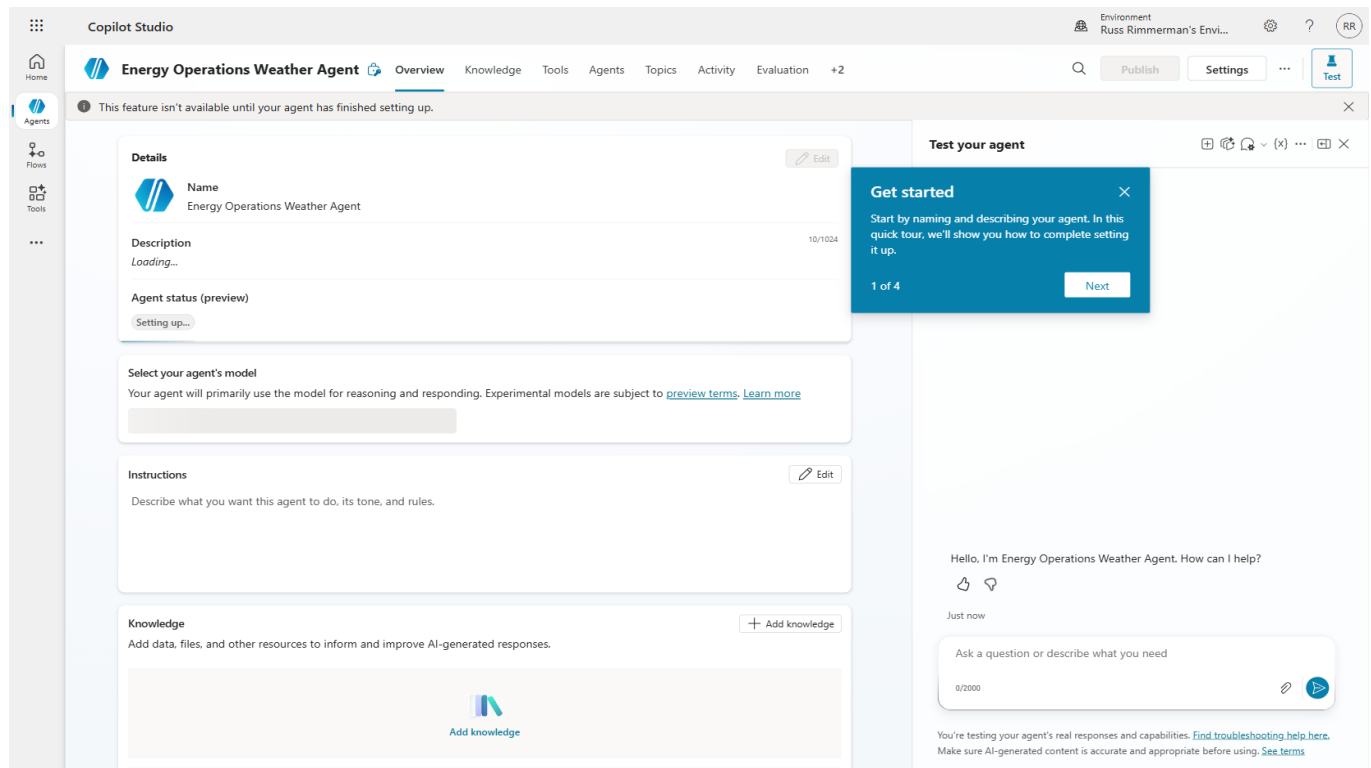
1.2 Configure Agent Identity

Step 3: In the “Name your agent” dialog, enter: Energy Operations Weather Agent. Add a description that explains its purpose for grid operations.



TIP: Choose descriptive names that clearly indicate the agent's domain. This helps with discoverability when you have multiple agents in your environment.

Step 4: After creation, you will see the Agent Overview page. This is your central hub for managing all aspects of the agent.



1.3 Add Agent Instructions

Instructions tell the AI model how to behave. For an energy operations agent, we need specific guidance about utility terminology, safety protocols, and response formatting.

Step 5: In the Instructions field, enter comprehensive guidance covering: role definition, weather analysis protocols, grid impact assessment criteria, and response formatting standards.

The screenshot displays the Copilot Studio interface for an agent named "Energy Operations Weather Agent". The interface is divided into two main sections: the left sidebar and the main content area.

Left Sidebar: Contains navigation icons for Home, Agents, Flows, Tools, and a menu icon.

Main Content Area: The "Overview" tab is selected. It contains the following sections:

- Details:** Shows the agent's name "Energy Operations Weather Agent", a description "None provided", and an agent status of "2 Warnings".
- Select your agent's model:** A dropdown menu shows "Claude Sonnet 4.6".
- Instructions:** A text area contains the following instructions: "You are a grid operations assistant for dispatchers, distribution planners, field crews, and customer-operations teams. Help users translate live weather and short-range forecasts into operational decisions about load, outages, crew staging, and customer impact. Ask for the city and state when location is missing or ambiguous. Default to Imperial units unless the user requests Metric. Keep responses concise, time-aware, and useful for grid operations — call out heat-driven AC peaks, cold-load pickup, severe-weather risk, and crew safety where relevant." The character count is 558/8000.
- Knowledge:** A section for adding data, files, and other resources to inform and improve AI-generated responses. It includes an "Add knowledge" button.

Test your agent panel: On the right, the "Test your agent" panel shows a chat interaction. The agent responds to the greeting "Hello, I'm Energy Operations Weather Agent. How can I help?" with a thumbs up and thumbs down icon. The chat history shows "2 minutes ago". Below the chat, there is a text input field with the placeholder "Ask a question or describe what you need" and a character count of 0/2000. At the bottom, a note states: "You're testing your agent's real responses and capabilities. Find troubleshooting help here. Make sure AI-generated content is accurate and appropriate before using. See terms".

⚠ NOTE: Instructions are critical for agent quality. They define the AI's persona, capabilities, and constraints. Well-written instructions prevent hallucination and ensure domain-appropriate responses.

Section 2: Create a Topic with Variables

Topics are conversational flows that define how your agent handles specific intents. Variables store user inputs and intermediate data as the conversation progresses. In this section, you will create a “Service Territory Weather Lookup” topic that collects city and state information from the user.

2.1 Navigate to Topics

Step 1: Click the “Topics” tab in the agent navigation bar. This shows all topics (system and custom) for your agent.

The screenshot shows the Copilot Studio interface for the 'Energy Operations Weather Agent'. The 'Topics' tab is selected in the navigation bar. The main area displays a table of custom topics:

Name	Type	Trigger	Last modified	Editing	Errors	Blocked	Enabled
Goodbye	Topic	By agent	Russ Rimmerman 3 min...				On
Greeting	Topic	By agent	Russ Rimmerman 3 min...				On
Start Over	Topic	By agent	Russ Rimmerman 3 min...				On
Thank you	Topic	By agent	Russ Rimmerman 3 min...				On

On the right, the 'Test your agent' chat window shows a greeting: 'Hello, I'm Energy Operations Weather Agent. How can I help?' with a timestamp of '3 minutes ago' and a text input field for the user to ask a question.

Step 2: Click “Add” → “From blank” to create a new custom topic.

Copilot Studio Environment: Russ Rimmerman's Envi... RR

Energy Operations Weather Agent Overview Knowledge Tools Agents **Topics** Activity Evaluation +2

Search custom topics Last refreshed now

Test your agent

Hello, I'm Energy Operations Weather Agent. How can I help?

4 minutes ago

Ask a question or describe what you need

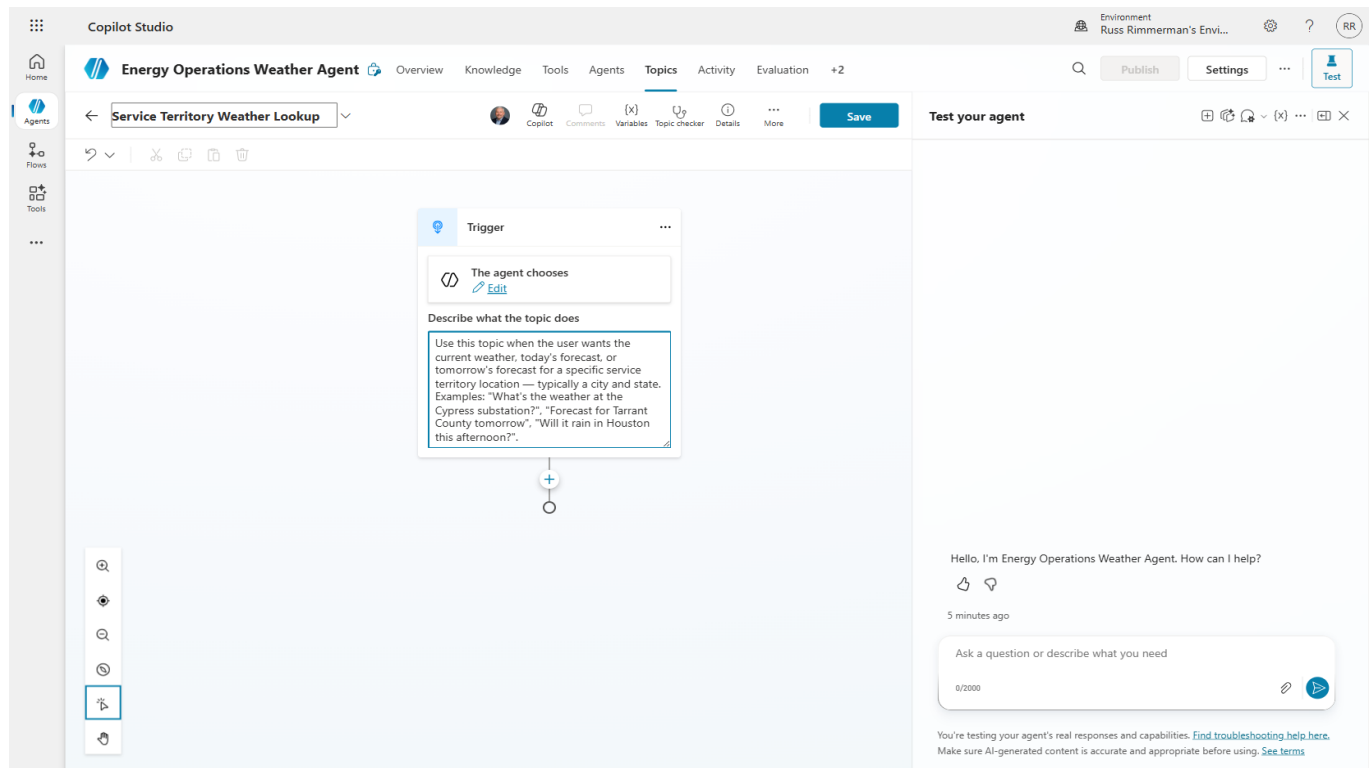
0/2000

You're testing your agent's real responses and capabilities. [Find troubleshooting help here.](#) Make sure AI-generated content is accurate and appropriate before using. [See terms](#)

Name	Type	Trigger	Last modified	Editing	Errors	Blocked	Enabled
Goodbye	Topic	By agent	Russ Rimmerman 3 min...				On
Greeting	Topic	By agent	Russ Rimmerman 3 min...				On
Start Over	Topic	By agent	Russ Rimmerman 3 min...				On
Thank you	Topic	By agent	Russ Rimmerman 3 min...				On

2.2 Configure Topic Details

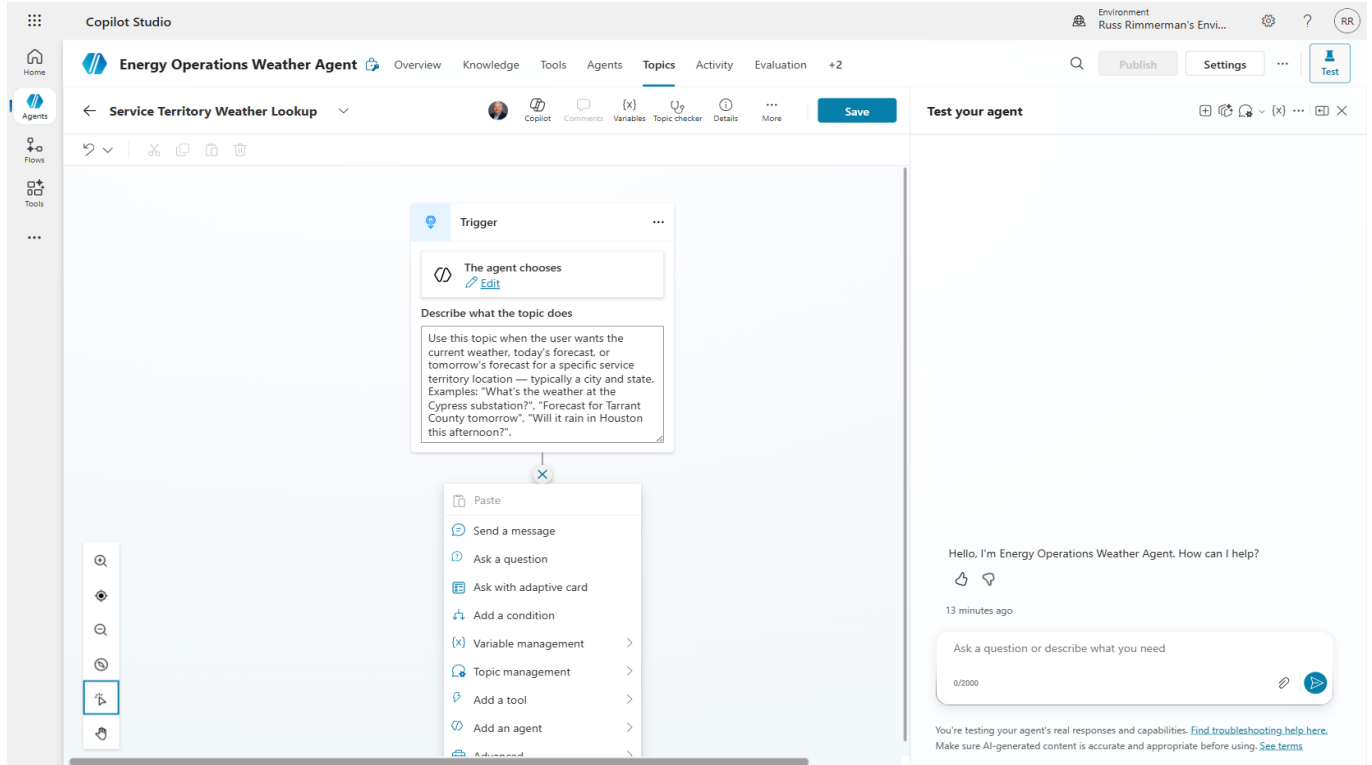
Step 3: Name the topic "Service Territory Weather Lookup" and add trigger phrases such as: "weather in", "current conditions", "forecast for", "grid weather".



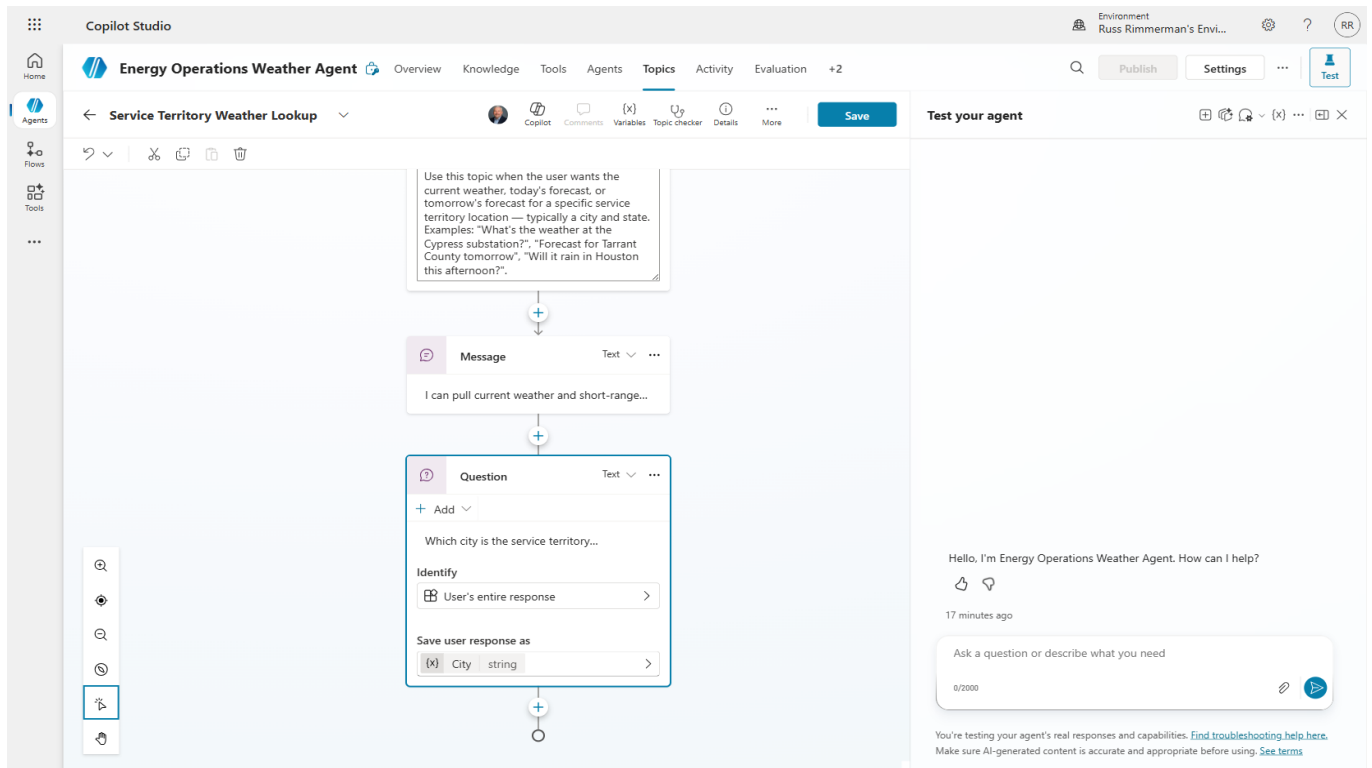
2.3 Add Question Nodes

Question nodes prompt the user for information and store their response in a variable. We need two pieces of data: city and state.

Step 4: Click the "+" button to add a node, then select "Ask a question."



Step 5: Configure the first question to ask “What city is the service territory located in?”. Set the response to save to a variable named “City”. Add a second question for State.



Step 6: Click "Save" to preserve your topic. You should see a "Topic Saved!" confirmation banner.

The screenshot shows the Copilot Studio interface for the 'Energy Operations Weather Agent'. The 'Topics' tab is selected, showing a flowchart with three steps: 'User's entire response', 'Question', and 'Message'. The 'Question' step is highlighted with a blue box. A green banner at the top left says 'Topic Saved!'. On the right, the 'Test your agent' panel shows a chat interface with a greeting and a text input field.

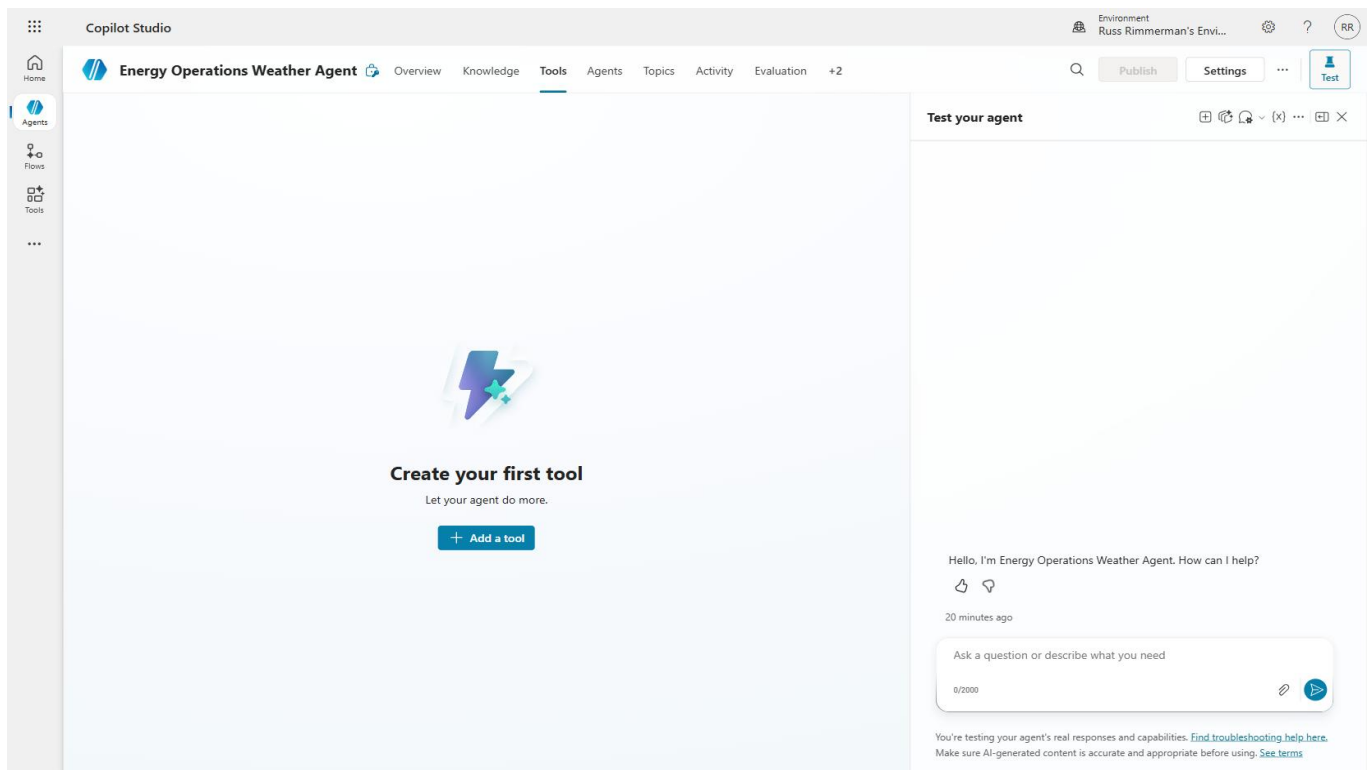
TIP: Always save your topic before navigating away. Unsaved changes will be lost.

Section 3: Add MSN Weather Connector Tools

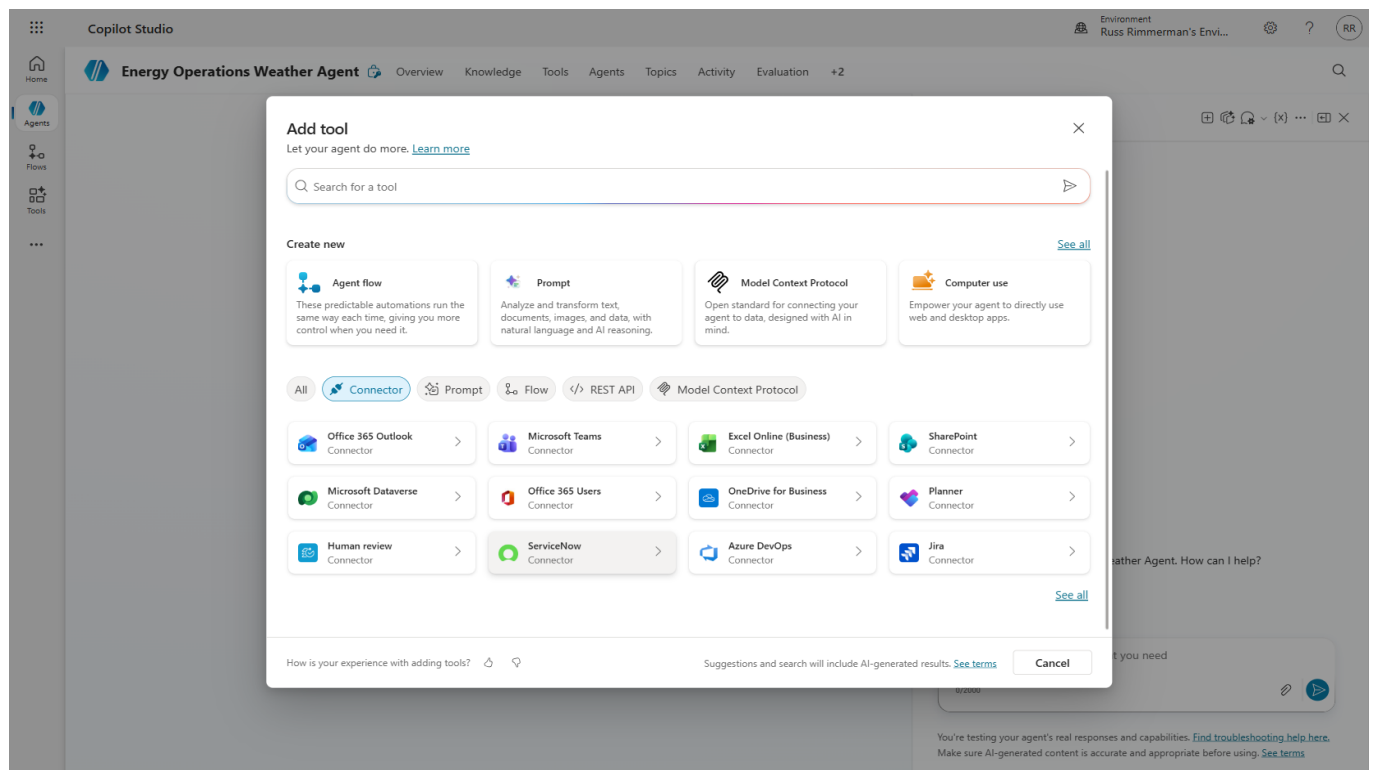
Tools connect your agent to external data sources and services. The MSN Weather connector provides real-time weather data including current conditions, today's forecast, and tomorrow's forecast. Adding multiple tools gives the agent flexibility to answer different types of weather questions.

3.1 Navigate to Tools

Step 1: Click the "Tools" tab. Initially, this will be empty since no tools have been added yet.

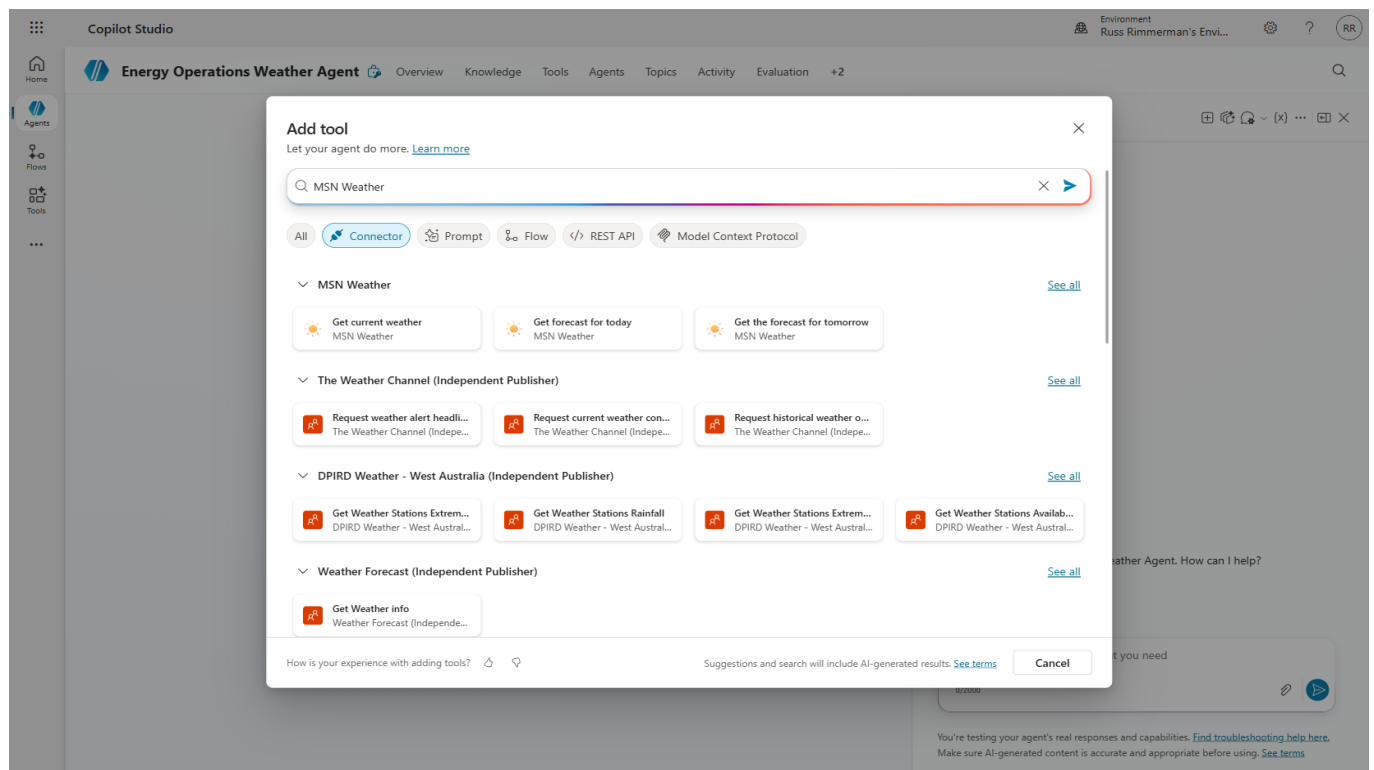


Step 2: Click "Add a tool" to open the tool gallery. This shows all available connectors, prompts, flows, and custom tools.

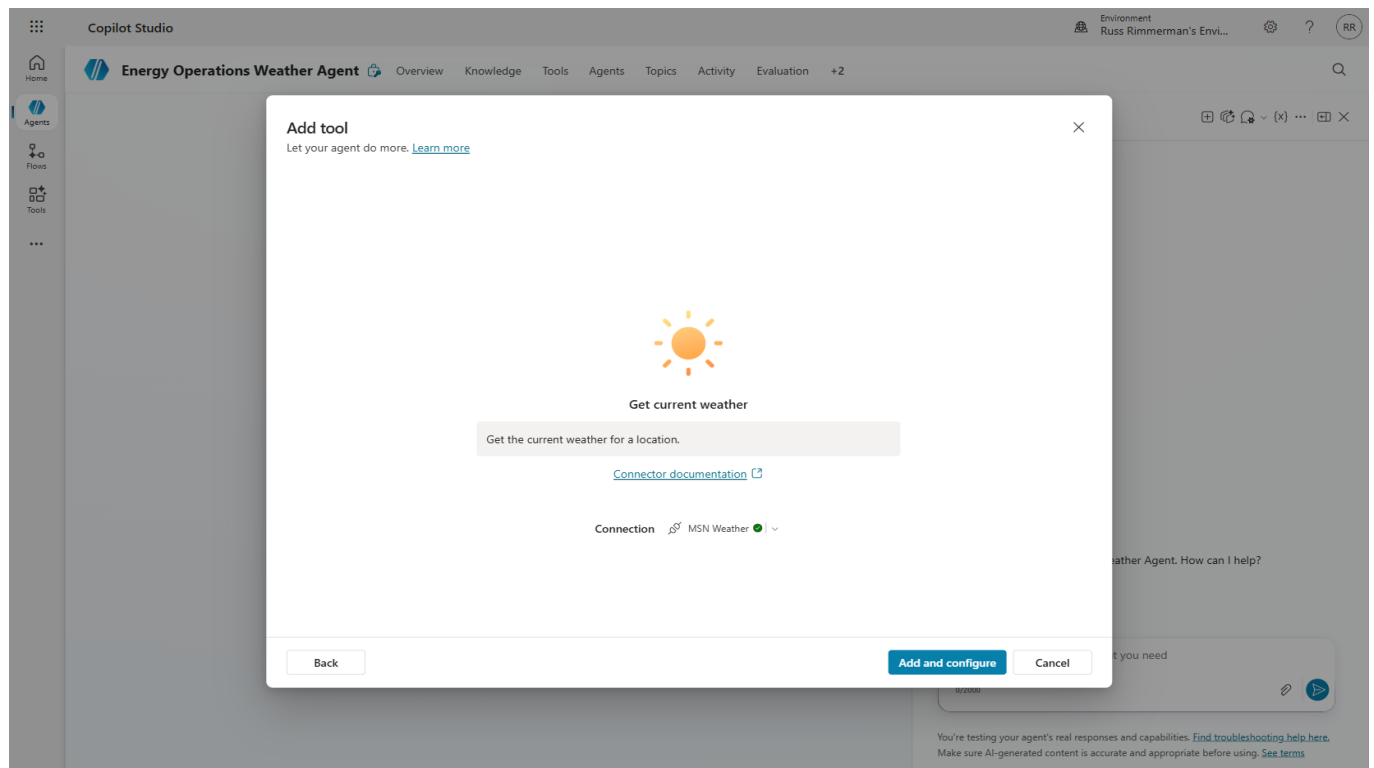


3.2 Add MSN Weather Connectors

Step 3: Search for “MSN Weather” in the tool gallery. You will see three available actions: Get current weather, Get forecast for today, and Get the forecast for tomorrow.



Step 4: Select “Get current weather” first. Review the tool details showing its inputs (Location, Units) and outputs (weather data object).



Step 5: Click “Add” to add the tool to your agent. Configure any required settings on the tool configuration page.

The screenshot shows the Copilot Studio interface for the 'Energy Operations Weather Agent'. The 'Tools' tab is selected, and the configuration page for the 'Get current weather' tool is displayed. The tool is currently 'Enabled'. The configuration includes sections for 'Details', 'Inputs', and 'Completion'. Under 'Inputs', there are fields for 'Location' and 'Units', both with 'Fill with AI' and 'Custom' options. A 'Test your agent' button is visible in the top right corner.

The screenshot shows the Copilot Studio interface for the 'Energy Operations Weather Agent'. The 'Tools' tab is selected, and the 'Tools' list is displayed. The 'Get current weather' tool is listed with its configuration details. The tool is currently 'Enabled'. The list includes columns for Name, Type, Available to, Trigger, Last modified, Errors, Blocked, and Enabled. A 'Test your agent' button is visible in the top right corner.

Name	Type	Available to	Trigger	Last modified	Errors	Blocked	Enabled
Get current weather	Connector	Energy Ops	By agent	Russ Rimmerman 42 se...			On

Step 6: Repeat the process for “Get forecast for today” and “Get the forecast for tomorrow.”

Lab 04: Energy Operations Weather Agent

The screenshot shows the Copilot Studio interface for the 'Energy Operations Weather Agent'. The 'Tools' tab is selected, displaying the configuration for the 'Get forecast for today' tool. The tool is enabled and has a 'Save' button. The configuration includes an 'Inputs' section with 'Location' and 'Units' fields, both with 'Fill with AI' and 'Custom' options. A 'Completion' section is also visible. On the right, the 'Test your agent' panel shows a chat interface with a greeting and a text input field.

Step 7: Verify all three tools appear in your Tools list.

The screenshot shows the 'Tools' list for the 'Energy Operations Weather Agent'. The 'Tools' tab is selected, and the 'Add a tool' button is visible. Below the button, there is a table listing the tools. The table has columns for Name, Type, Available to, Trigger, Last modified, Errors, Blocked, and Enabled. Three tools are listed: 'Get current weather', 'Get forecast for tod...', and 'Get the forecast for...'. All three tools are enabled and have a 'By agent' trigger.

Name	Type	Available to	Trigger	Last modified	Errors	Blocked	Enabled
Get current weather	Connector	Energy Opi	By agent	Russ Rimmerman 4 min...			On
Get forecast for tod...	Connector	Energy Opi	By agent	Russ Rimmerman 2 min...			On
Get the forecast for...	Connector	Energy Opi	By agent	Russ Rimmerman 32 se...			On

NOTE: Adding all three weather tools gives the orchestrator options. It can intelligently select which tool to call based on the user's question (current vs. forecast).

Section 4: Wire Tools into Topic with Power Fx

Power Fx is Microsoft's low-code formula language (similar to Excel). In Copilot Studio, Power Fx connects your topic variables to tool inputs and formats tool outputs for display. This is where your conversational flow becomes a functioning data pipeline.

4.1 Add Tool Node to Topic

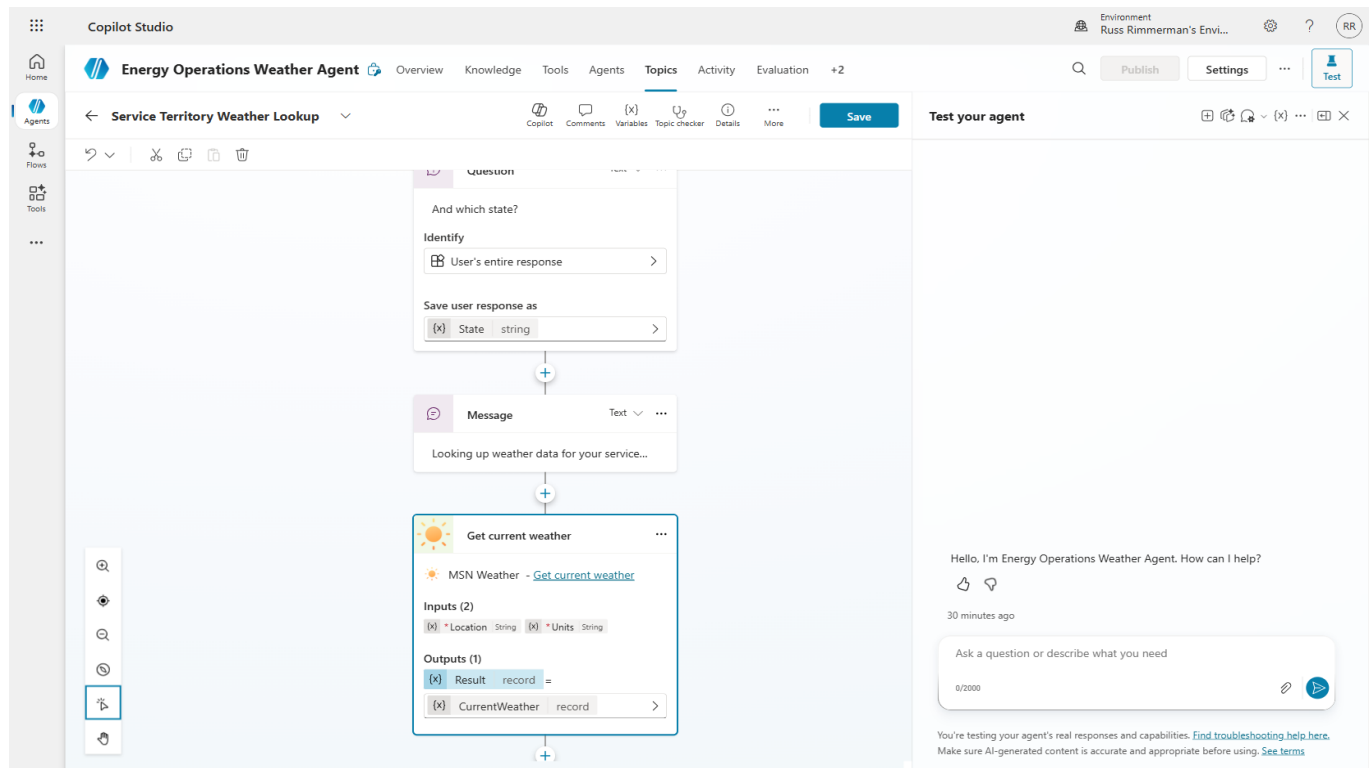
Step 1: Return to the "Service Territory Weather Lookup" topic. Your current flow has the two question nodes. Click "+" after the State question to add a new node.

The screenshot displays the Copilot Studio interface for the "Energy Operations Weather Agent". The main workspace shows the "Service Territory Weather Lookup" topic flow. The flow is composed of three nodes connected sequentially:

- Identify Node:** Input is "User's entire response", output is "City" (string).
- Question Node:** Question is "And which state?", input is "User's entire response", output is "State" (string).
- Message Node:** Message is "Looking up weather data for your..."

The "Add" button is highlighted in the bottom left corner of the flow editor. The right panel shows a chat preview with the agent's greeting: "Hello, I'm Energy Operations Weather Agent. How can I help?". Below the greeting is a text input field with the placeholder "Ask a question or describe what you need".

Step 2: Select "Call an action" and choose the "Get current weather" tool from the list.



4.2 Configure Power Fx Input Formulas

The weather tool requires a Location input (city, state format) and Units (Imperial/Metric). We use Power Fx to combine our topic variables into the correct format.

Step 3: Click the Location input field and enter the Power Fx formula: `Topic.City & ", " & Topic.State`

The screenshot shows the Copilot Studio interface for the 'Energy Operations Weather Agent'. The 'Topics' tab is selected, and the 'Get current weather' topic is being edited. The 'Inputs' tab is active, showing two inputs: 'Location' (String) and 'Units' (String). The 'Outputs' tab shows two outputs: 'Result' (record) and 'CurrentWeather' (CurrentWeather). The 'Enter formula' dialog is open, showing the formula field with the text 'Topic.City & \", \" & Topic.State'. The dropdown menu is open, showing 'State' selected. The 'Type' is 'String' and the 'Output' is 'Topic.City & \", \" & Topic.State'. The background shows the 'Get current weather' topic configuration with inputs for Location and Units, and outputs for Result and CurrentWeather.

Step 4: Set the Units input to the formula: "Imperial" (hardcoded for US energy operations).

The screenshot shows the Copilot Studio interface for the 'Energy Operations Weather Agent'. The 'Topics' tab is selected, and the 'Get current weather' topic is being edited. The 'Inputs' tab is active, showing two inputs: 'Location' (String) and 'Units' (String). The 'Units' input is now set to the formula 'Imperial'. The 'Outputs' tab shows two outputs: 'Result' (record) and 'CurrentWeather' (CurrentWeather). The background shows the 'Get current weather' topic configuration with inputs for Location and Units, and outputs for Result and CurrentWeather.

4.3 Add Response Message

After the tool executes, we need to display results to the user. The tool returns a complex object with nested weather properties.

Step 5: Add a Message node after the tool node. Use Power Fx expressions to format the response, referencing properties like `Topic.CurrentWeather.responses.weather.current.temp`.

The screenshot shows the Copilot Studio interface for the 'Energy Operations Weather Agent'. The topic flow is titled 'Service Territory Weather Lookup'. It consists of three nodes: a 'Message' node with the text 'Looking up weather data for your service...', a 'Get current weather' tool node, and another 'Message' node. The 'Get current weather' tool node has inputs for 'Location' (String) and 'Units' (String), and outputs for 'Result' (record) and 'CurrentWeather' (record). An 'Enter formula' dialog is open, showing the formula: `Text(Topic.CurrentWeather.responses.weather.current.temp) & '°F'`. The dialog also shows the output type as 'String' and the output as 'Text(Topic.CurrentWeather...)'. The background shows the topic flow and a chat interface with a message: 'Hello, I'm Energy Operations Weather Agent. How can I help?'.

Step 6: Review the complete topic flow showing: Questions → Tool Call → Response Message.

The screenshot displays the Copilot Studio interface for the 'Energy Operations Weather Agent'. The main workspace shows a flowchart with three steps:

- Question:** Asks 'And which state?'. It identifies the user's entire response and saves it as 'State' (string).
- Message:** Says 'Looking up weather data for your service...'.
- Get current weather:** Uses the 'MSDN Weather' tool. Inputs are 'Location' (string) and 'Units' (string). The output is 'Result' (record), which is recorded as 'CurrentWeather'.

The right panel shows a chat window with the agent's greeting: 'Hello, I'm Energy Operations Weather Agent. How can I help?'. Below the chat, there is a test input field with the text 'Ask a question or describe what you need'.

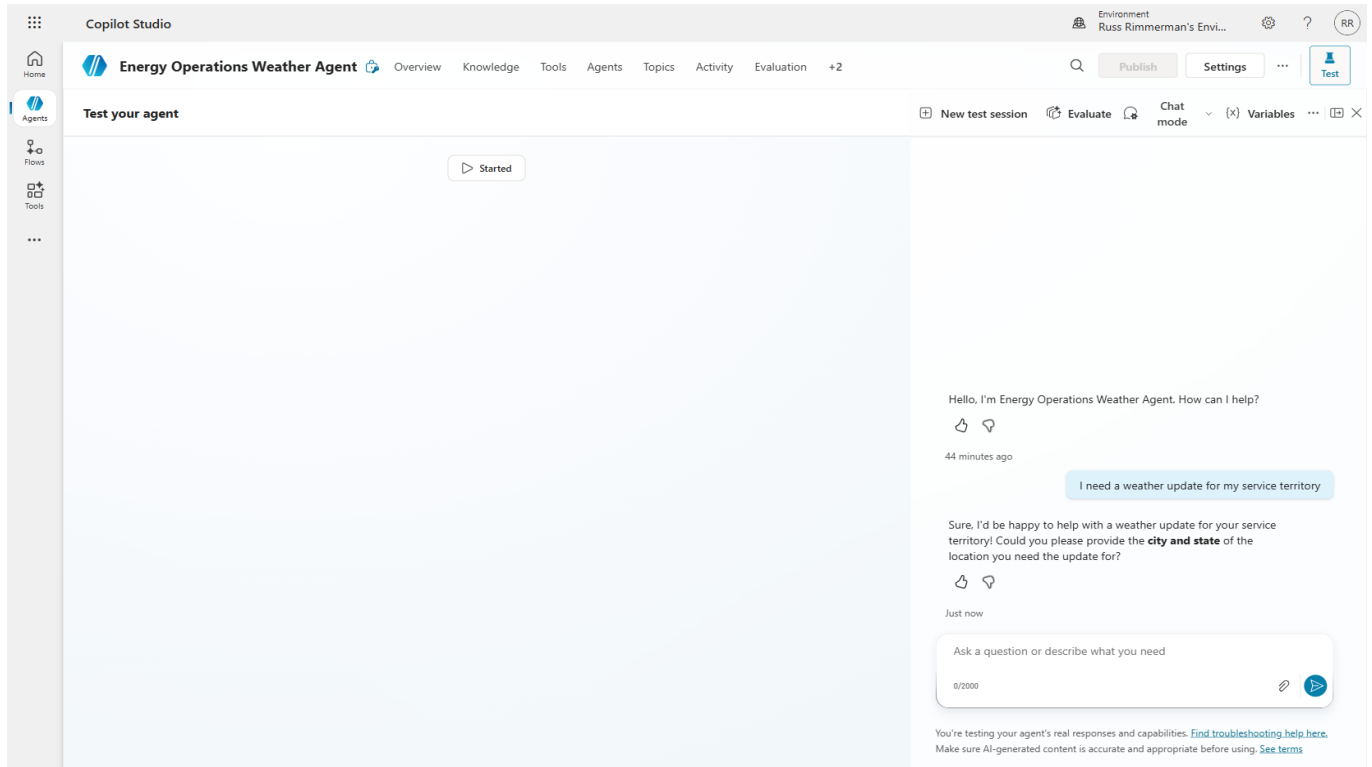
TIP: Power Fx formulas are validated in real-time. A green checkmark means the formula is syntactically correct. Red errors indicate type mismatches or invalid property references.

Section 5: Test the Agent

Testing validates that your entire flow works end-to-end: the agent triggers the correct topic, collects user inputs, calls the weather API with the right parameters, and returns formatted results. The built-in Test panel simulates a real conversation.

5.1 Open Test Panel

Step 1: Click the “Test” button in the upper right corner to open the test chat panel.



5.2 Authenticate Connector

The first time a connector is used, it requires authentication. This is a one-time step per connector per user.

Step 2: When prompted to connect, click the connection link to authenticate with MSN Weather.

Lab 04: Energy Operations Weather Agent

Copilot Studio Environment: Russ Rimmerman's Envi... ? RR

Energy Operations Weather Agent Overview Knowledge Tools Agents Topics Activity Evaluation +2

Test your agent New test session Evaluate Chat mode (x) Variables

Get current weather Connector Action

Get forecast for today Connector Action

Get current weather Working

Description
Get the current weather for a location.

Inputs
current_weather_location (String)
Dallas, TX, USA
current_weather_units (String)
I

Outputs
Latitude (Number)
Longitude (Number)
Location (String)
Pressure (Number)
Conditions (String)

[Edit tool](#) [Documentation](#)

Hello, I'm Energy Operations Weather Agent. How can I help?

44 minutes ago

I need a weather update for my service territory

Sure, I'd be happy to help with a weather update for your service territory! Could you please provide the **city and state** of the location you need the update for?

Just now

Dallas

Let's get you connected first, and then I can find that info for you. [Open connection manager](#) to verify your credentials.

Once the connection is ready, retry your request.

[Retry](#) [Cancel](#)

Just now

Ask a question or describe what you need

9/2000

You're testing your agent's real responses and capabilities. [Find troubleshooting help here.](#) Make sure AI-generated content is accurate and appropriate before using. [See terms](#)

Manage your connections [Sign out](#)

Manage connections Refresh

All Connected Deactivated Expired Not Connected Stale

	Name	Used By	Status
☐	MSN Weather MSN Weather gets you the very latest weather forecast, including temperature, humidity, p...	3 tools	Not Connected Connect

Step 3: After authenticating, verify the connection shows as established.

Manage your connections[Sign out](#)

Manage connections

Refresh

All

Connected

Deactivated

Expired

Not Connected

Stale

☐	Name	Used By	Status
☐	 MSN Weather MSN Weather gets you the very latest weather forecast, including temperature, humidity, p...	3 tools	Connected Manage

5.3 End-to-End Test

Step 4: Type a weather query such as “What’s the weather in Dallas, Texas?” The agent should ask for city/state (or extract from your message), call the weather API, and return live conditions.

Copilot Studio

Environment: Russ Rimmerman's Envi...

Energy Operations Weather Agent Overview Knowledge Tools Agents Topics Activity Evaluation +2

Test your agent

Get current weather Connector Action 143.34s

Get forecast for today Connector Action 0.36s

Completed

Get forecast for today Connector Action Completed

Description: Get the forecast for the current day in the specified location.

Inputs:

- current_weather_location (String): Dallas, TX, USA
- current_weather_units (String): I

Outputs:

- Moon Phase (String): Waxing Crescent
- Moon Phase Code (String): WxCr
- Moonrise Time (DateTime):
- Moonsset Time (DateTime):
- Sunrise Time (DateTime):

Operational Highlights:

- AC Peak Load Risk — HIGH:** High of 94°F with a 75°F dew point will push heat index values well above 100°F during afternoon hours. Expect significant demand surge between 2 PM – 8 PM CDT. Monitor for transformer loading and substation capacity.
- Wind Near Threshold:** Daytime winds at 17 mph are approaching the 17.7 mph operational threshold — monitor for gusty conditions and potential line movement.
- Crew Safety:** Extreme heat conditions during afternoon work windows. Enforce hydration protocols, limit outdoor exposure during peak heat hours, and consider scheduling non-critical field work for early morning.
- UV Very High:** Ensure field crews have appropriate sun protection.
- Precipitation:** Minimal risk today — no widespread outage threat from storms.

High / Low: 94°F 77°F

Wind: 17 mph SSE 15 mph SSE

Precip. Chance: 13% 5%

UV Index: 8 – Very High Low

Ask a question or describe what you need

0/2000

You're testing your agent's real responses and capabilities. [Find troubleshooting help here.](#) Make sure AI-generated content is accurate and appropriate before using. [See terms](#)

Copilot Studio

Environment: Russ Rimmerman's Envi...

Energy Operations Weather Agent Overview Knowledge Tools Agents Topics Activity Evaluation +2

Test your agent

Get current weather Connector Action 143.34s

Get forecast for today Connector Action 0.36s

Completed

Get forecast for today Connector Action Completed

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High / Low: 94°F 77°F

Wind: 17 mph SSE 15 mph SSE

Precip. Chance: 13% 5%

UV Index: 8 – Very High Low

Ask a question or describe what you need

0/2000

You're testing your agent's real responses and capabilities. [Find troubleshooting help here.](#) Make sure AI-generated content is accurate and appropriate before using. [See terms](#)

Energy Operations Weather Agent Overview Knowledge Tools Agents Topics Activity Evaluation +2

Test your agent

Get current weather Connector Action 143.34s

Get forecast for today Connector Action 0.36s

Completed

High / Low	94°F	77°F
Wind	17 mph SSE	15 mph SSE
Precip. Chance	13%	5%
UV Index	8 – Very High	Low

Operational Highlights

- 🔥 **AC Peak Load Risk — HIGH:** High of 94°F with a 75°F dew point will push heat index values well above 100°F during afternoon hours. Expect significant demand surge between **2 PM – 8 PM CDT**. Monitor for transformer loading and substation capacity.
- 🌬️ **Wind Near Threshold:** Daytime winds at 17 mph are approaching the 17.7 mph operational threshold — monitor for gusty conditions and potential line movement.
- 👷 **Crew Safety:** Extreme heat conditions during afternoon work windows. Enforce hydration protocols, limit outdoor exposure during peak heat hours, and consider scheduling non-critical field work for early morning.
- ☀️ **UV Very High:** Ensure field crews have appropriate sun protection.
- 🌧️ **Precipitation:** Minimal risk today — no widespread outage threat from storms.

2 minutes ago

Ask a question or describe what you need

0/2000

You're testing your agent's real responses and capabilities. [Find troubleshooting help here.](#) Make sure AI-generated content is accurate and appropriate before using. [See terms](#)

⚠️ **NOTE:** Live weather data means your results will differ from these screenshots. That's expected! Verify the structure (temperature, conditions, humidity, wind) is correct, not the specific values.

Section 6: Create a Custom Prompt Tool

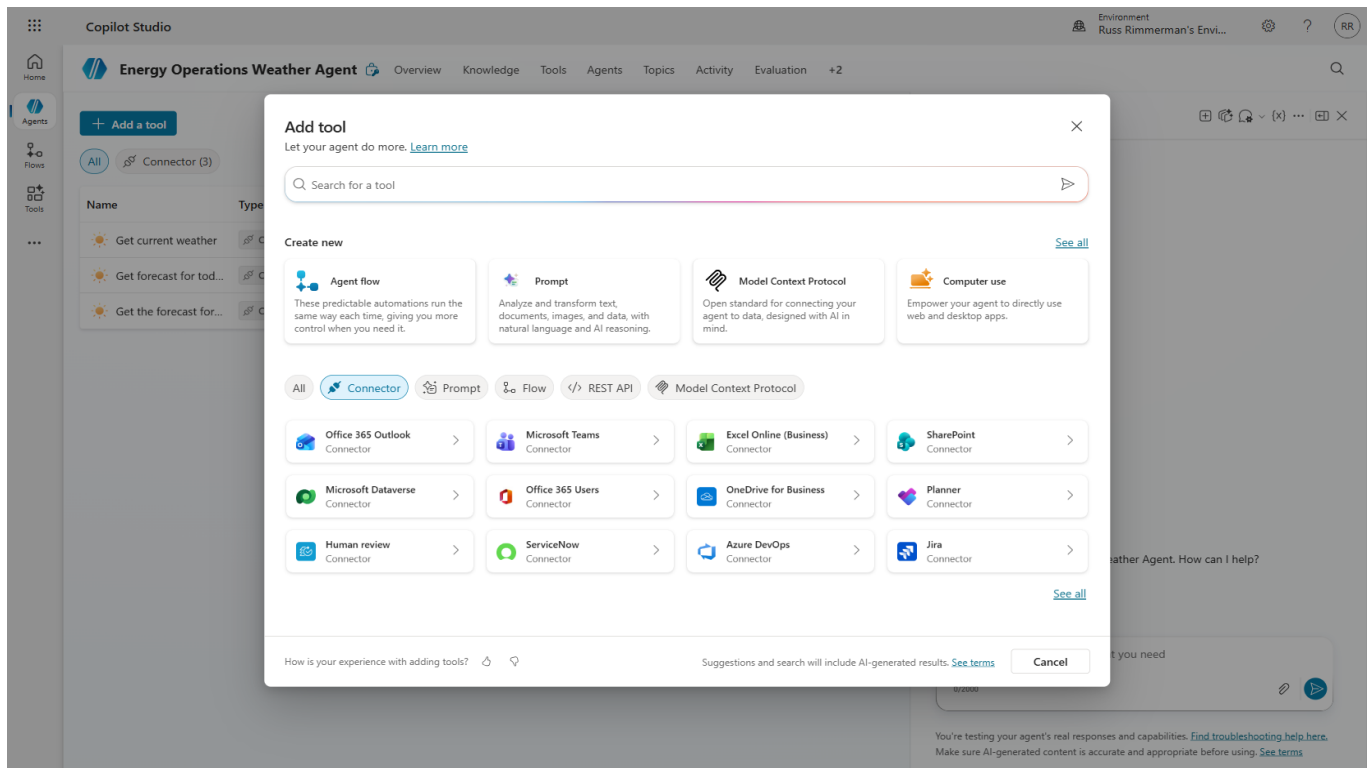
Custom Prompt Tools use generative AI to transform data into insights. Instead of just showing raw weather data, we will create a prompt that acts as an “Energy Operations Analyst” — taking weather data and generating a structured operations briefing with grid impact analysis, demand forecasts, and recommended actions.

Why this matters:

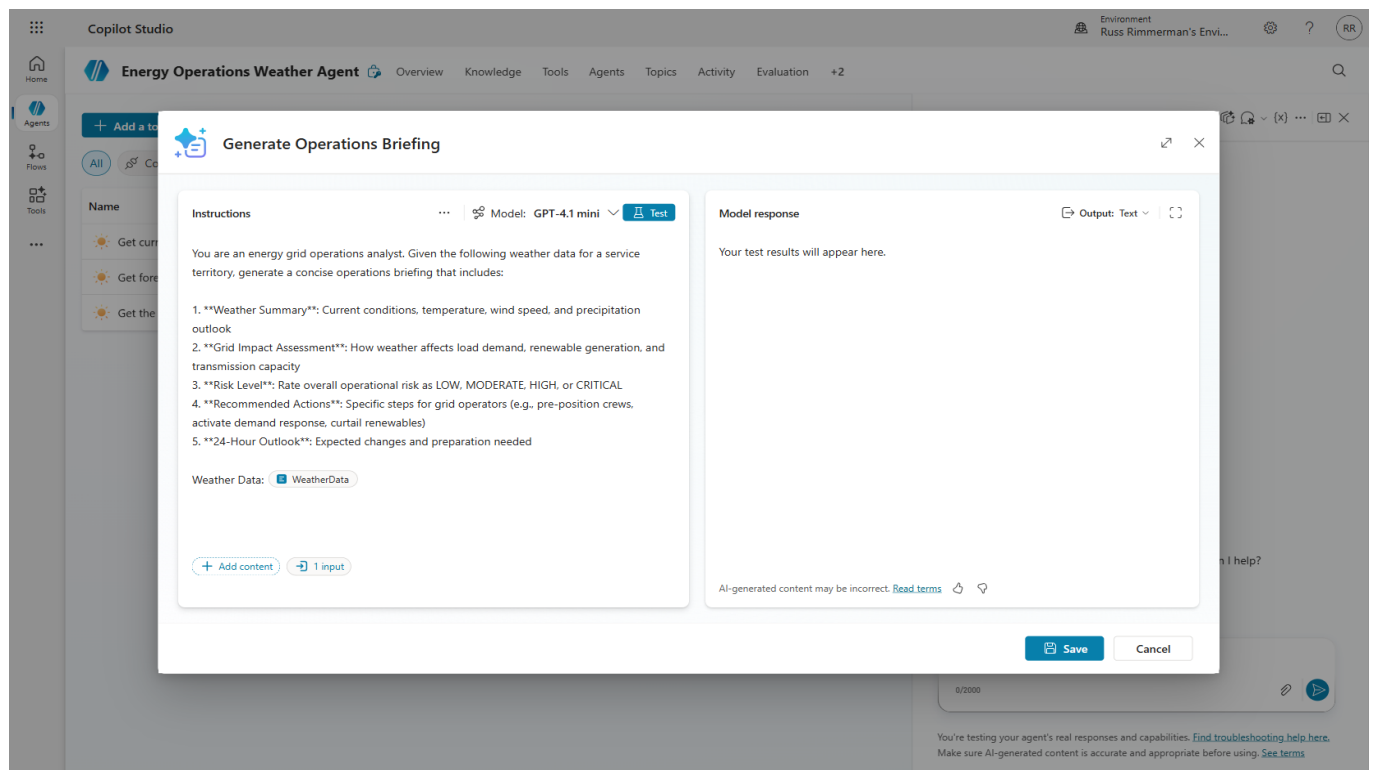
Raw data requires human interpretation. A prompt tool automates the analysis step, turning a temperature reading into an actionable “peak demand expected, pre-position crews” recommendation.

6.1 Create the Prompt

Step 1: Navigate to Tools → “Add a tool” and select “Prompt” from the “Create new” section.



Step 2: Name the prompt “Generate Operations Briefing” and enter detailed instructions that define the AI’s role as an energy operations analyst. Include sections for: Grid Operations Implications, Active Weather Alerts analysis, Demand Forecast, and Recommended Actions.



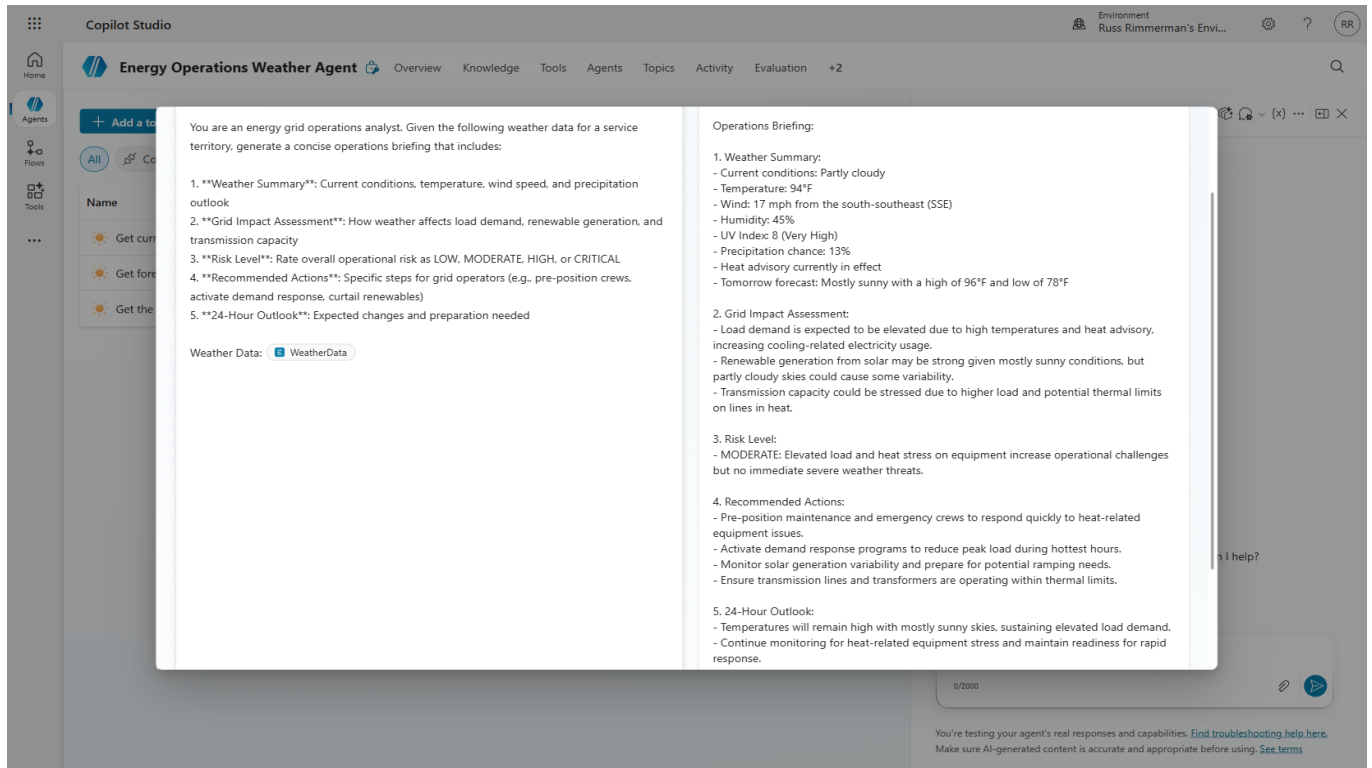
6.2 Add Input Variable

The prompt needs weather data to analyze. We create an input variable that will receive the formatted weather data from the connector output.

Step 3: Add a text input variable called "WeatherData" with sample data for testing the prompt independently.

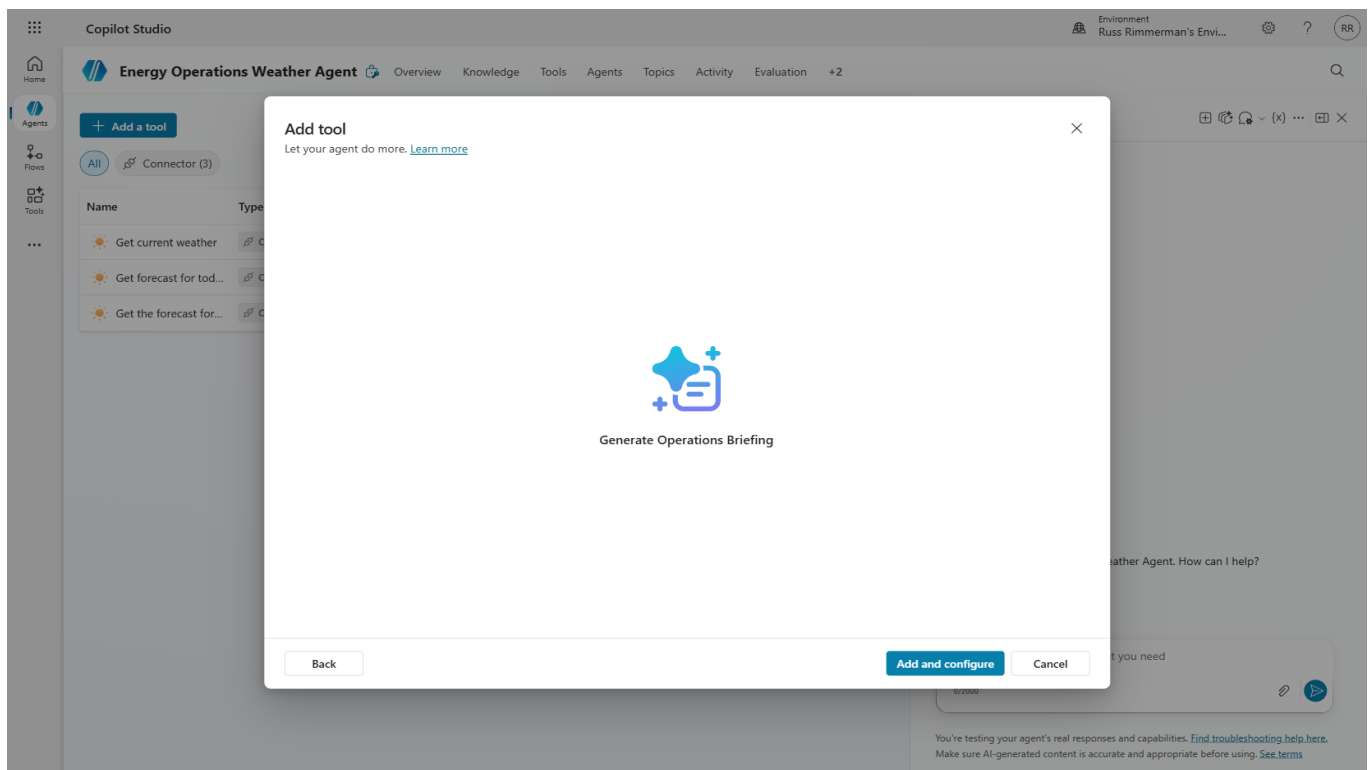
6.3 Test the Prompt

Step 4: Click "Test prompt" to validate the AI generates a proper operations briefing from the sample data.



TIP: Always test prompts independently before wiring them into a topic. This isolates prompt engineering issues from flow logic issues.

Step 5: Save the prompt and add it to the agent's tool set.



Lab 04: Energy Operations Weather Agent

The screenshot displays the Copilot Studio interface for the 'Energy Operations Weather Agent'. The top navigation bar includes 'Home', 'Agents', 'Flows', and 'Tools'. The main header shows the agent's name and tabs for 'Overview', 'Knowledge', 'Tools', 'Agents', 'Topics', 'Activity', and 'Evaluation'. The 'Tools' tab is active, showing the 'Generate Operations Briefing' tool. This tool is configured with a name, description, and is available to the 'Energy Operations Weather Agent'. The instruction section defines the agent's role as an energy grid operations analyst and provides a list of tasks: generating a weather summary and a grid impact assessment. The inputs section lists 'WeatherData' as a required input. On the right, the 'Test your agent' panel shows a chat interface with a greeting and a prompt to ask a question or describe a need.

Copilot Studio Environment: Russ Rimmerman's Envi... Settings Test

Energy Operations Weather Agent Overview Knowledge **Tools** Agents Topics Activity Evaluation +2

Generate Operations Briefing Enabled Save

Details
What it is, how it operates, and how the orchestrator identifies it. [Learn more](#)
Name *
Generate Operations Briefing 28/64
Description * ⓘ
Generates an energy grid operations briefing from weather data, including risk levels, grid impact assessment, and recommended actions for operators. 149/1024
Available to
Energy Operations Weather Agent
[Additional details](#)

Instruction GPT-4.1 mini ⓘ
You are an energy grid operations analyst. Given the following weather data for a service territory, generate a concise operations briefing that includes:
1. **Weather Summary**: Current conditions, temperature, wind speed, and precipitation outlook
2. **Grid Impact Assessment**: How weather affects load demand, renewable generation, and transmission capacity
[Add content](#)

Inputs + Add input
What the tool accepts in order to run. Inputs will be filled in the order shown.
WeatherData *
AI will fill this in for you, no action is needed Fill with AI Custom

Test your agent Test

Hello, I'm Energy Operations Weather Agent. How can I help?
20 minutes ago
Ask a question or describe what you need
0/2000

You're testing your agent's real responses and capabilities. [Find troubleshooting help here.](#)
Make sure AI-generated content is accurate and appropriate before using. [See terms](#)

Section 7: Wire Prompt Tool into Topic Flow

Now we integrate the prompt tool into our existing topic flow. The data pipeline becomes: User Input → Weather API Call → AI Analysis (Prompt) → Operations Briefing. This creates a two-stage AI pipeline where the first stage retrieves data and the second stage generates intelligence.

7.1 Add Prompt Node

Step 1: Open the "Service Territory Weather Lookup" topic. Add the "Generate Operations Briefing" tool node between the weather connector output and the response message.

7.2 Configure Input Formula

Step 2: Set the WeatherData input using a Power Fx formula that concatenates relevant fields from the weather connector output:

"Location: " & Topic.City & ", " & Topic.State & " | Conditions: " & Topic.CurrentWeather.responses.weather.current.sky & " | Temperature: " & Text(Topic.CurrentWeather.responses.weather.current.temp) & "°F"

The screenshot shows the Copilot Studio interface for the "Energy Operations Weather Agent". The "Topics" tab is selected, and the "Service Territory Weather Lookup" topic is open. The flow diagram shows a sequence of nodes: a "Message" node, a "Get current weather" node, and a "Prompt" node. The "Get current weather" node has two inputs: "Location" (String) and "Units" (String). The "Prompt" node has one input: "WeatherData" (String). A formula editor is open, showing the formula for the "WeatherData" input:

```
fx CurrentWeather.responses.
weather.current.rh) & "% |
Wind: " & Topic.
CurrentWeather.responses.
weather.current.windSpd & "
mph
```

The formula editor also shows the "Type" as "String" and the "Output" as "Location: " & Topic.City".

7.3 Update Response Message

Step 3: Modify the final message node to display the OperationsBriefing.text output from the prompt tool.

The screenshot shows the Copilot Studio interface for the 'Energy Operations Weather Agent'. The 'Topics' tab is selected, and the 'Service Territory Weather Lookup' topic is being edited. The flowchart shows a sequence of nodes: a 'WeatherData' node, a 'Generate Operations Briefing' node, and a 'Message' node. The 'Message' node is currently being edited, and the 'OperationsBriefing.text' output from the 'Generate Operations Briefing' node is being assigned to the message content. The right-hand pane shows a chat interface with the agent's greeting: 'Hello, I'm Energy Operations Weather Agent. How can I help?'.

Step 4: Save the topic.

The screenshot shows the Copilot Studio interface for the 'Energy Operations Weather Agent'. The 'Topics' tab is selected, and the 'Service Territory Weather Lookup' topic is being edited. The flowchart shows a sequence of nodes: a 'Get current weather' node, a 'Prompt' node, and a 'Generate Operations Briefing' node. The 'Prompt' node is currently being edited, and the 'OperationsBriefing.text' output from the 'Generate Operations Briefing' node is being assigned to the prompt content. The right-hand pane shows a chat interface with the agent's greeting: 'Hello, I'm Energy Operations Weather Agent. How can I help?'.

7.4 Test Full Pipeline

Step 5: Test the complete flow. You should now receive a comprehensive operations briefing including weather conditions, grid impact analysis, and recommended actions.

Copilot Studio Environment: Russ Rimmerman's Envi...

Energy Operations Weather Agent Overview Knowledge Tools Agents Topics Activity Evaluation +2

Test your agent

Get current weather Connector Action 0.49s Completed

Get current weather Completed Connector Action

Description
Get the current weather for a location.

Inputs

- current_weather_units (String): I
- current_weather_location (String): Houston, TX, USA

Outputs

- Latitude (Number): 29.763280868530273
- Longitude (Number): -95.36327362060547
- Location (String): Houston, TX
- Pressure (Number): 29.62
- Conditions (String):

[Edit tool](#) [Documentation](#)

Active Weather Alerts

- Heat Advisory** — Upcoming Effective today at 11:00 AM CDT through June 19 at 10:00 PM CDT "Prepare for possible heat conditions. Pay close attention to weather forecasts and alerts."
- Tropical Cyclone Statement** — Ongoing Issued by the National Weather Service, valid through 3:30 AM CDT today

Grid Operations Implications

- Overnight baseline is elevated** — 80°F with 90% humidity and an 85°F feels-like at midnight is well above seasonal norms. Overnight load relief will be minimal.
- Heat Advisory kicks in this afternoon** — Expect a **strong AC-driven peak load** during afternoon hours. Prepare for demand surge, especially in residential feeders.
- Tropical Cyclone activity** — Monitor for any residual wind or moisture impacts from the active tropical cyclone statement. Field crew deployments should factor in evolving conditions.
- Crew safety** — High overnight heat indices signal dangerous conditions for field personnel; ensure hydration protocols and heat exposure limits are enforced for any crews working today.

Would you like today's or tomorrow's forecast for Houston to support load planning?

Just now

Ask a question or describe what you need

0/2000

You're testing your agent's real responses and capabilities. [Find troubleshooting help here.](#) Make sure AI-generated content is accurate and appropriate before using. [See terms](#)

Copilot Studio

Environment: Russ Rimmerman's Envi...

Energy Operations Weather Agent Overview Knowledge Tools Agents Topics Activity Evaluation +2

Test your agent

Get current weather Connector Action 0.49s

Completed

Get current weather Connector Action Completed

Description

Get the current weather for a location.

Inputs

current_weather_units (String)

I

current_weather_location (String)

Houston, TX, USA

Outputs

Latitude (Number)

29.763280868530273

Longitude (Number)

-95.36327362060547

Location (String)

Houston, TX

Pressure (Number)

29.62

Conditions (String)

Edit tool Documentation

Pressure 29.62 in

Air Quality Good (AQI 43, PM2.5)

Active Weather Alerts

Heat Advisory — Upcoming Effective today at 11:00 AM CDT through June 19 at 10:00 PM CDT "Prepare for possible heat conditions. Pay close attention to weather forecasts and alerts."

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Just now

Ask a question or describe what you need

0/2000

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Copilot Studio

Environment: Russ Rimmerman's Envi...

Energy Operations Weather Agent Overview Knowledge Tools Agents Topics Activity Evaluation +2

Test your agent

Get current weather Connector Action 0.49s

Completed

Pressure 29.62 in

Air Quality Good (AQI 43, PM2.5)

Active Weather Alerts

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- Crew safety — High overnight heat indices signal dangerous conditions for field personnel: ensure hydration protocols and heat exposure limits are enforced for any crews working today.

Would you like today's or tomorrow's forecast for Houston to support load planning?

2 minutes ago

Ask a question or describe what you need

0/2000

You're testing your agent's real responses and capabilities. [Find troubleshooting help here.](#) Make sure AI-generated content is accurate and appropriate before using. [See terms](#)

NOTE: The full pipeline (API call + AI analysis) typically takes 6-10 seconds. This is expected for real-time data retrieval plus generative AI processing.

Section 8: Connected Agents (Child Agent)

Connected Agents allow you to decompose complex capabilities into specialized sub-agents. The parent agent's orchestrator automatically routes conversations to child agents based on the user's intent. This follows the microservices pattern — each agent has a focused responsibility.

Why use connected agents:

- Separation of concerns — each agent has focused instructions
- Independent testing and iteration
- Reusability across multiple parent agents
- Cleaner orchestration at scale

8.1 Navigate to Agents Tab

Step 1: Click the "Agents" tab to see connected agent management.

The screenshot shows the Copilot Studio interface for the 'Energy Operations Weather Agent'. The 'Agents' tab is selected in the top navigation bar. The main workspace displays a large 'Add an agent' button with the text 'They'll work together to complete your workflow.' Below this, there is a small '+ Add' button. The right sidebar, titled 'Test your agent', contains a table with weather data and a list of active weather alerts.

Parameter	Value
Pressure	29.62 in
Air Quality	Good (AQI 43, PM2.5)

Active Weather Alerts

- Heat Advisory** — Upcoming Effective today at 11:00 AM CDT through June 19 at 10:00 PM CDT "Prepare for possible heat conditions. Pay close attention to weather forecasts and alerts."
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Grid Operations Implications

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- Crew safety** — High overnight heat indices signal dangerous conditions for field personnel; ensure hydration protocols and heat exposure limits are enforced for any crews working today.

Would you like today's or tomorrow's forecast for Houston to support load planning?

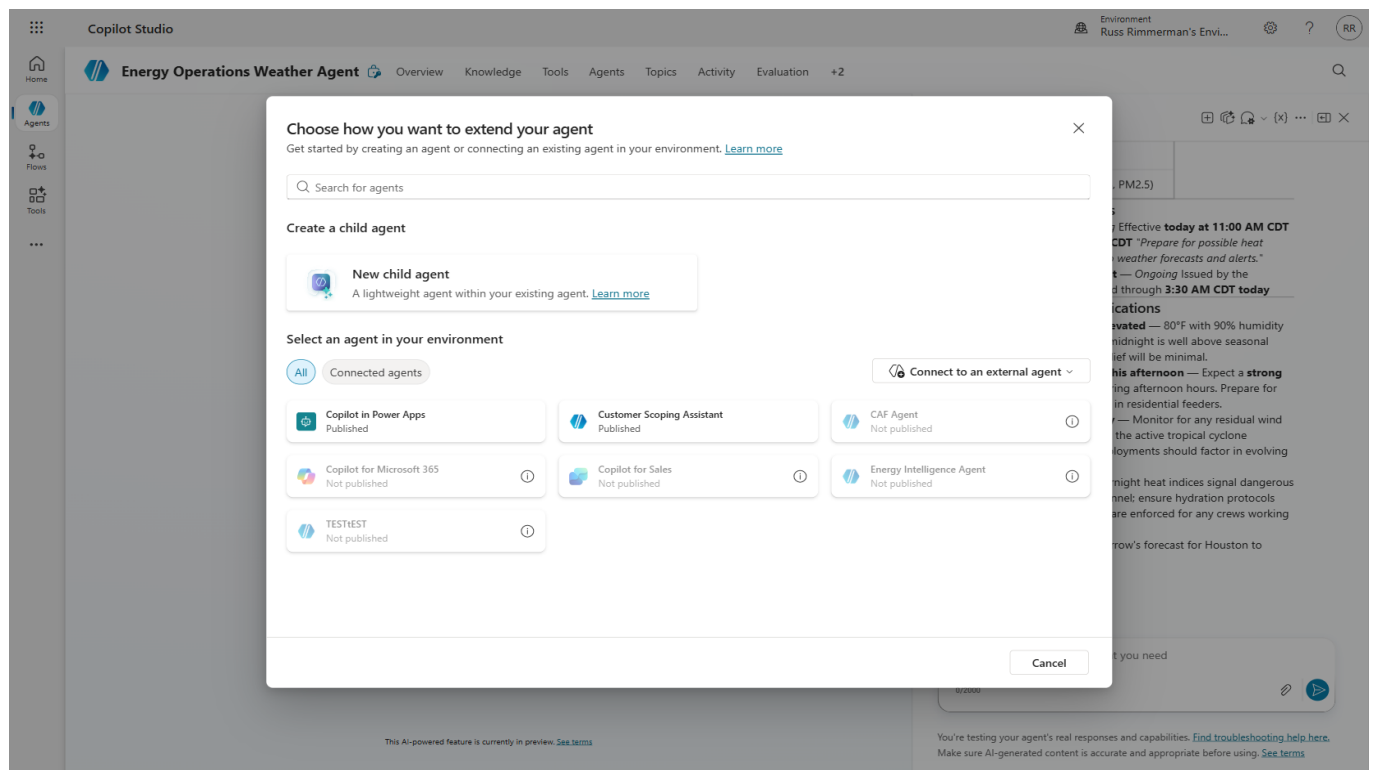
3 minutes ago

Ask a question or describe what you need

0/2000

You're testing your agent's real responses and capabilities. [Find troubleshooting help here.](#) Make sure AI-generated content is accurate and appropriate before using. [See terms](#)

Step 2: Click "Add" to see the options for extending your agent.



8.2 Create Child Agent

Step 3: Select “New child agent.” Configure the Crew Safety Advisor with a name, description, and detailed instructions covering WBGT (Wet Bulb Globe Temperature) methodology, work/rest cycles, storm protocols, PPE adjustments, and mandatory stand-down conditions.

Lab 04: Energy Operations Weather Agent

Copilot Studio Environment: Russ Rimmerman's Envi...

Energy Operations Weather Agent Overview Knowledge Tools **Agents** Topics Activity Evaluation +2

Crew Safety Advisor Enabled **Save**

Details

What it is, how it operates, and how the orchestrator identifies it. [Learn more](#)

Name
Crew Safety Advisor

When will this be used?
The agent chooses - Based on description

Description *
Enter text
Missing required property 'Description'

Advanced

You are a Crew Safety Advisor for energy utility field operations. When given weather data or conditions for a service territory:

1. Assess heat stress risk using WBGT (Wet Bulb Globe Temperature) methodology
2. Determine appropriate work/rest cycles based on heat index
3. Identify storm-related safety protocols (lightning, high winds, flooding)
4. Recommend PPE adjustments for weather conditions
5. Flag any mandatory stand-down conditions

Always provide:

- A safety risk level (Green/Yellow/Orange/Red)
- Specific recommended work/rest ratios
- Required safety briefing topics for the shift
- Any mandatory crew notifications

Be concise, authoritative, and prioritize crew safety above operational targets.

Test your agent

Pressure: 29.62 in

Air Quality: Good (AQI 43, PM2.5)

Active Weather Alerts

- Heat Advisory** — Upcoming Effective today at 11:00 AM CDT through June 19 at 10:00 PM CDT "Prepare for possible heat conditions. Pay close attention to weather forecasts and alerts."
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Grid Operations Implications

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- Tropical Cyclone activity** — Monitor for any residual wind or moisture impacts from the active tropical cyclone statement. Field crew deployments should factor in evolving conditions.
- Crew safety** — High overnight heat indices signal dangerous conditions for field personnel; ensure hydration protocols and heat exposure limits are enforced for any crews working today.

Would you like today's or tomorrow's forecast for Houston to support load planning?

5 minutes ago

Ask a question or describe what you need

9/2000

You're testing your agent's real responses and capabilities. [Find troubleshooting help here.](#) Make sure AI-generated content is accurate and appropriate before using. [See terms](#)

Step 4: Save the child agent.

Copilot Studio Environment: Russ Rimmerman's Envi...

Energy Operations Weather Agent Overview Knowledge Tools **Agents** Topics Activity Evaluation +2

Crew Safety Advisor Enabled **Save**

Details

What it is, how it operates, and how the orchestrator identifies it. [Learn more](#)

Name
Crew Safety Advisor

When will this be used?
The agent chooses - Based on description

Description *
Evaluates weather conditions and provides crew safety recommendations including heat protocols, storm precautions.

Advanced

You are a Crew Safety Advisor for energy utility field operations. When given weather data or conditions for a service territory:

1. Assess heat stress risk using WBGT (Wet Bulb Globe Temperature) methodology
2. Determine appropriate work/rest cycles based on heat index
3. Identify storm-related safety protocols (lightning, high winds, flooding)
4. Recommend PPE adjustments for weather conditions
5. Flag any mandatory stand-down conditions

Always provide:

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- Required safety briefing topics for the shift
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Be concise, authoritative, and prioritize crew safety above operational targets.

Test your agent

Pressure: 29.62 in

Air Quality: Good (AQI 43, PM2.5)

Active Weather Alerts

- Heat Advisory** — Upcoming Effective today at 11:00 AM CDT through June 19 at 10:00 PM CDT "Prepare for possible heat conditions. Pay close attention to weather forecasts and alerts."
- Tropical Cyclone Statement** — Ongoing Issued by the National Weather Service, valid through 3:30 AM CDT today

Grid Operations Implications

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- Crew safety** — High overnight heat indices signal dangerous conditions for field personnel; ensure hydration protocols and heat exposure limits are enforced for any crews working today.

Would you like today's or tomorrow's forecast for Houston to support load planning?

6 minutes ago

Ask a question or describe what you need

9/2000

You're testing your agent's real responses and capabilities. [Find troubleshooting help here.](#) Make sure AI-generated content is accurate and appropriate before using. [See terms](#)

Step 5: Verify the Crew Safety Advisor appears in the Agents tab with "Child" relationship, "By agent" trigger, and "Enabled: On."

The screenshot displays the Copilot Studio interface for the 'Energy Operations Weather Agent'. The main panel shows a table of agents, with 'Crew Safety Advisor' listed as a child agent. The right-hand 'Test your agent' panel shows a live test of the agent's capabilities, including weather data and active alerts.

Name	Relationship	Trigger	Last modified	Errors	Blocked	Enabled
Crew Safety Advisor	Child	By agent	Russ Rimmerman 21 56...			On

Test your agent

Environment: Russ Rimmerman's Envi...

Pressure: 29.62 in

Air Quality: Good (AQI 43, PM2.5)

Active Weather Alerts

- Heat Advisory** — Upcoming Effective today at 11:00 AM CDT through June 19 at 10:00 PM CDT "Prepare for possible heat conditions. Pay close attention to weather forecasts and alerts."
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Grid Operations Implications

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- Heat Advisory kicks in this afternoon** — Expect a **strong AC-driven peak load** during afternoon hours. Prepare for demand surge, especially in residential feeders.
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- Crew safety** — High overnight heat indices signal dangerous conditions for field personnel; ensure hydration protocols and heat exposure limits are enforced for any crews working today.

Would you like today's or tomorrow's forecast for Houston to support load planning?

7 minutes ago

Ask a question or describe what you need

0/2000

This AI-powered feature is currently in preview. [See terms](#)

You're testing your agent's real responses and capabilities. [Find troubleshooting help here.](#) Make sure AI-generated content is accurate and appropriate before using. [See terms](#)

8.3 Test Connected Agent Orchestration

The parent agent's orchestrator will automatically route crew safety questions to the child agent based on intent matching.

Step 6: Ask a crew safety question such as "Are there any crew safety concerns due to the heat in Houston today?" Watch the activity map to see the orchestrator hand off to the Crew Safety Advisor.

Copilot Studio Environment: Russ Rimmerman's Envi...

Energy Operations Weather Agent Overview Knowledge Tools Agents Topics Activity Evaluation +2

Test your agent

Get current weather Connector Action 0.49s

Get forecast for today Connector Action 0.36s

Crew Safety Advisor Agent

Search sources Knowledge

Search sources Knowledge

Search sources Knowledge

Total 3 Show less

Working

Search sources Knowledge Completed

Query

NIOSH ACGIH WBGT work rest ratio table heat index 91 to 103 degrees outdoor heavy work utility workers

Referenced sources

Heat Stress & WBGT Calculator ...
 "Heat Stress & WBGT Calculator Home calculator Heat Stress & WBGT Calculator by Jawad Chand - March 16, 2026 Heat Stress & WBGT Calculator WBGT Calculator Work/Rest Regimen Metabolic Rate Heat Index ..."

Heat Stress Work/Rest Schedules - Heat.gov ...
 "Heat Stress Work/Rest Schedules - Heat.gov Heat Stress Work/Rest Schedules This document provides a sample work and rest schedule for workers wearing normal clothing to ensure they have adequate rest from heat ..."

Heat Stress: Work/Rest Schedules ...
 "Heat Stress: Work/Rest Schedules heat stress Work/Rest Schedules Using work/rest schedules can decrease the risk of heat illness Sample Work/Rest Schedule for Workers Wearing Normal Clothing" The NIOSH work/re..."

Rationale ⓘ
[Show rationale](#)

[Edit knowledge](#)

Tropical Cyclone Statement — Ongoing Issued by the National Weather Service, valid through 3:30 AM CDT today

Grid Operations Implications

- Overnight baseline is elevated** — 80°F with 90% humidity and an 85°F feels-like at midnight is well above seasonal norms. Overnight load relief will be minimal.
- Heat Advisory kicks in this afternoon** — Expect a **strong AC-driven peak load** during afternoon hours. Prepare for demand surge, especially in residential feeders.
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- Crew safety** — High overnight heat indices signal dangerous conditions for field personnel; ensure hydration protocols and heat exposure limits are enforced for any crews working today.

Would you like today's or tomorrow's forecast for Houston to support load planning?

7 minutes ago

What are the crew safety recommendations for Dallas today given the heat advisory?

Searching through knowledge sources...

Ask a question or describe what you need

0/2000

You're testing your agent's real responses and capabilities. [Find troubleshooting help here.](#) Make sure AI-generated content is accurate and appropriate before using. [See terms](#)

Copilot Studio Environment: Russ Rimmerman's Envi...

Energy Operations Weather Agent Overview Knowledge Tools Agents Topics Activity Evaluation +2

Test your agent

Get current weather Connector Action 0.49s

Get forecast for today Connector Action 0.36s

Crew Safety Advisor Agent

Search sources Knowledge

Search sources Knowledge

Search sources Knowledge

Total 3 Show less

Working

Search sources Knowledge Completed

Query

NIOSH ACGIH WBGT work rest ratio table heat index 91 to 103 degrees outdoor heavy work utility workers

Referenced sources

Heat Stress & WBGT Calculator ...
 "Heat Stress & WBGT Calculator Home calculator Heat Stress & WBGT Calculator by Jawad Chand - March 16, 2026 Heat Stress & WBGT Calculator WBGT Calculator Work/Rest Regimen Metabolic Rate Heat Index ..."

Heat Stress Work/Rest Schedules - Heat.gov ...
 "Heat Stress Work/Rest Schedules - Heat.gov Heat Stress Work/Rest Schedules This document provides a sample work and rest schedule for workers wearing normal clothing to ensure they have adequate rest from heat ..."

Heat Stress: Work/Rest Schedules ...
 "Heat Stress: Work/Rest Schedules heat stress Work/Rest Schedules Using work/rest schedules can decrease the risk of heat illness Sample Work/Rest Schedule for Workers Wearing Normal Clothing" The NIOSH work/re..."

Rationale ⓘ
[Show rationale](#)

[Edit knowledge](#)

Tropical Cyclone Statement — Ongoing Issued by the National Weather Service, valid through 3:30 AM CDT today

Grid Operations Implications

- Overnight baseline is elevated** — 80°F with 90% humidity and an 85°F feels-like at midnight is well above seasonal norms. Overnight load relief will be minimal.
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Would you like today's or tomorrow's forecast for Houston to support load planning?

8 minutes ago

What are the crew safety recommendations for Dallas today given the heat advisory?

Searching through knowledge sources...

Ask a question or describe what you need

0/2000

You're testing your agent's real responses and capabilities. [Find troubleshooting help here.](#) Make sure AI-generated content is accurate and appropriate before using. [See terms](#)

Copilot Studio Environment: Russ Rimmerman's Envi...

Energy Operations Weather Agent Overview Knowledge Tools Agents Topics Activity Evaluation +2

Test your agent

- Get current weather (Connector Action) 0.49s
- Get forecast for today (Connector Action) 0.36s
- Crew Safety Advisor (Agent) 69.65s**
 - Search sources Knowledge
 - Search sources Knowledge
 - Search sources Knowledge

Total 3 Show less

Crew Safety Advisor Agent Completed

Description
Evaluates weather conditions and provides crew safety recommendations including heat protocols, storm precautions, and work/rest schedules for field personnel.

Inputs
Task (String)
Evaluate crew safety recommendations for field personnel working in Dallas, TX today, June 18, 2026, given the following weather forecast:
 - Condition: Partly Sunny
 - High Temperature: 94°F
 - Low Temperature: 77°F
 - Humidity: 51%–84% (high overnight)
[Show more](#)

Outputs
Summary (String)
User requested crew safety recommendations for Dallas, TX on June 18, 2026 under an active Heat Advisory. Weather data retrieved: High 94°F, humidity 51–84%, wind 17 mph SSE, UV Index 8 (Very High), partly sunny, overnight low 77°F. WBGT-based analysis performed using NIOSH/ACGIH/OSHA guidelines. Overall safety risk level assigned: ORANGE. Key recommendations included: work/rest cycles (25/35 for heavy work / 45/15 for moderate / 45/15 for light) based on...

[Edit agent](#)

Crew Notifications Required

- All field supervisors notified of Orange-level heat protocol
- Dispatch notified — build extra time into job estimates for rest cycles
- Medical/EMS contact pre-briefed for territory
- Crew leads to confirm hydration supplies loaded before departure
- Document all heat safety briefings in **daily safety log**

Operational targets do not override crew safety. No task completion timeline justifies bypassing heat protocols under an active Heat Advisory.

4 references

- Heat - Heat Hazard Recog...
- Heat Stress: Work/Rest Sc...
- Heat Stress & WBGT Calcu...
- Workplace Recommendati...

Just now

Ask a question or describe what you need

0/2000

You're testing your agent's real responses and capabilities. [Find troubleshooting help here.](#) Make sure AI-generated content is accurate and appropriate before using. [See terms](#)

Copilot Studio Environment: Russ Rimmerman's Envi...

Energy Operations Weather Agent Overview Knowledge Tools Agents Topics Activity Evaluation +2

Test your agent

- Get current weather (Connector Action) 0.49s
- Get forecast for today (Connector Action) 0.36s
- Crew Safety Advisor (Agent) 69.65s**
 - Search sources Knowledge
 - Search sources Knowledge
 - Search sources Knowledge

Total 3 Show less

Crew Safety Advisor Agent Completed

Description
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Inputs
Task (String)
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[Show more](#)

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[Edit agent](#)

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4 references

- Heat - Heat Hazard Recog...
- Heat Stress: Work/Rest Sc...
- Heat Stress & WBGT Calcu...
- Workplace Recommendati...

9 minutes ago

Ask a question or describe what you need

0/2000

You're testing your agent's real responses and capabilities. [Find troubleshooting help here.](#) Make sure AI-generated content is accurate and appropriate before using. [See terms](#)

TIP: Connected agent orchestration can take 30-60 seconds as the child agent may perform its own knowledge searches and reasoning. This is normal for complex multi-agent scenarios.

Section 9: Agent Evaluation

Evaluations measure your agent's quality at scale by running multiple test cases and scoring responses against quality criteria. This is essential for production readiness — you need confidence that the agent handles diverse inputs correctly before deploying to users.

9.1 Create Test Set

Step 1: Navigate to the “Evaluation” tab.

The screenshot shows the Copilot Studio interface for the 'Energy Operations Weather Agent'. The 'Evaluation' tab is selected in the top navigation bar. The main content area features a large blue bell icon and the text 'Evaluate your agent's performance'. Below this, it says 'Make testing your agent easier and more comprehensive with large batches of questions. [Learn more](#)'. A blue button labeled 'Create a test set' is prominently displayed.

The right sidebar, titled 'Test your agent', contains a table with test cases and a list of references. The table has two columns: 'Test Case' and 'Expected Result'. The test cases are:

Test Case	Expected Result
WBGT on-site meter reads ≥ 30.5°C (heavy work)	Restrict to 25% work / 75% rest or suspend non-critical tasks
Heat Advisory escalates to Heat Warning	Reassess all active crews — suspend non-emergency work

Below the table, there is a section titled 'Crew Notifications Required' with a list of bullet points:

- **All field supervisors** notified of Orange-level heat protocol
- **Dispatch** notified — build extra time into job estimates for rest cycles
- **Medical/EMS contact** pre-briefed for territory
- **Crew leads** to confirm hydration supplies loaded before departure
- Document all heat safety briefings in **daily safety log**

Below the list, there is a note: 'Operational targets do not override crew safety. No task completion timeline justifies bypassing heat protocols under an active Heat Advisory.'

At the bottom of the sidebar, there is a section titled '4 references' with a list of four references:

- 1 Heat - Heat Hazard Recog...
- 2 Heat Stress: Work/Rest Sc...
- 3 Heat Stress & WBGT Calcu...
- 4 Workplace Recommendati...

Below the references, there is a text input field with the placeholder 'Ask a question or describe what you need' and a '0/2000' character count. A 'Test' button is visible next to the input field.

Step 2: Click “Create a test set.” Choose “Quick question set” to have the AI automatically generate relevant test questions based on your agent’s capabilities.

Copilot Studio Environment: Russ Rimmerman's Envi... ? RR

Energy Operations Weather Agent Overview Knowledge Tools Agents Topics Activity **Evaluation** +2

Evaluation > New evaluation

Data type

- ☒ **Single response**
Check how the agent responds to a specific question. Best for targeted capability testing.
- ☐ **Conversation (preview)**
Check the quality of longer agent interactions with users. Best for general assessments.

Data source

Start by uploading some questions
Use our [CSV](#) template to make sure that the file contents and formatting are correct. [Learn more](#)

Drag a CSV file or [browse](#)

You can still review and change things before starting the evaluation.

More ways to start

Quick question set
Generate 10 questions based on your agent's description, instructions and capabilities.

Full question set
Generate up to 100 questions, using knowledge or topics as a reference.

Use your test chat conversation
Gather the inputs and responses from your current manual testing session.

[Or, write some questions yourself](#)

Step 3: Review the AI-generated test questions. These cover weather lookups, grid operations analysis, crew safety, and cold-load pickup scenarios.

Copilot Studio Environment: Russ Rimmerman's Envi... ? RR

Energy Operations Weather Agent Overview Knowledge Tools Agents Topics Activity **Evaluation** +2

Evaluation > New evaluation

Review your test cases (10) AI-generated content may be incorrect

+ Add questions

Question	Expected response
Can you give me the current weather conditions for Dallas, TX?	
What's the forecast for tomorrow in Chicago, IL?	
How will today's weather impact grid operations in Los Angeles?	
Is there a risk of severe weather affecting our service territory this we...	
Can you generate an operations briefing for Houston based on the la...	
What's the expected load impact from the heat wave in Phoenix?	
Are there any crew safety concerns due to storms in Miami today?	
How should we stage crews for potential outages in New York given t...	
Can you provide a weather lookup for our service territory in Denver?	
What's the cold-load pickup risk for Minneapolis tonight?	

Configure test set

Data type: Single response

Name *
Evaluate Energy Operations Weather Agent

Test method *
Criteria for checking this data. Select up to five methods. [Learn more](#)

- General quality**
Answers meet quality standards, such as relevance and c...

+ Add test method

User profile
We recommend using an authenticated profile with all the required connections, so that you can test all agent capabilities.

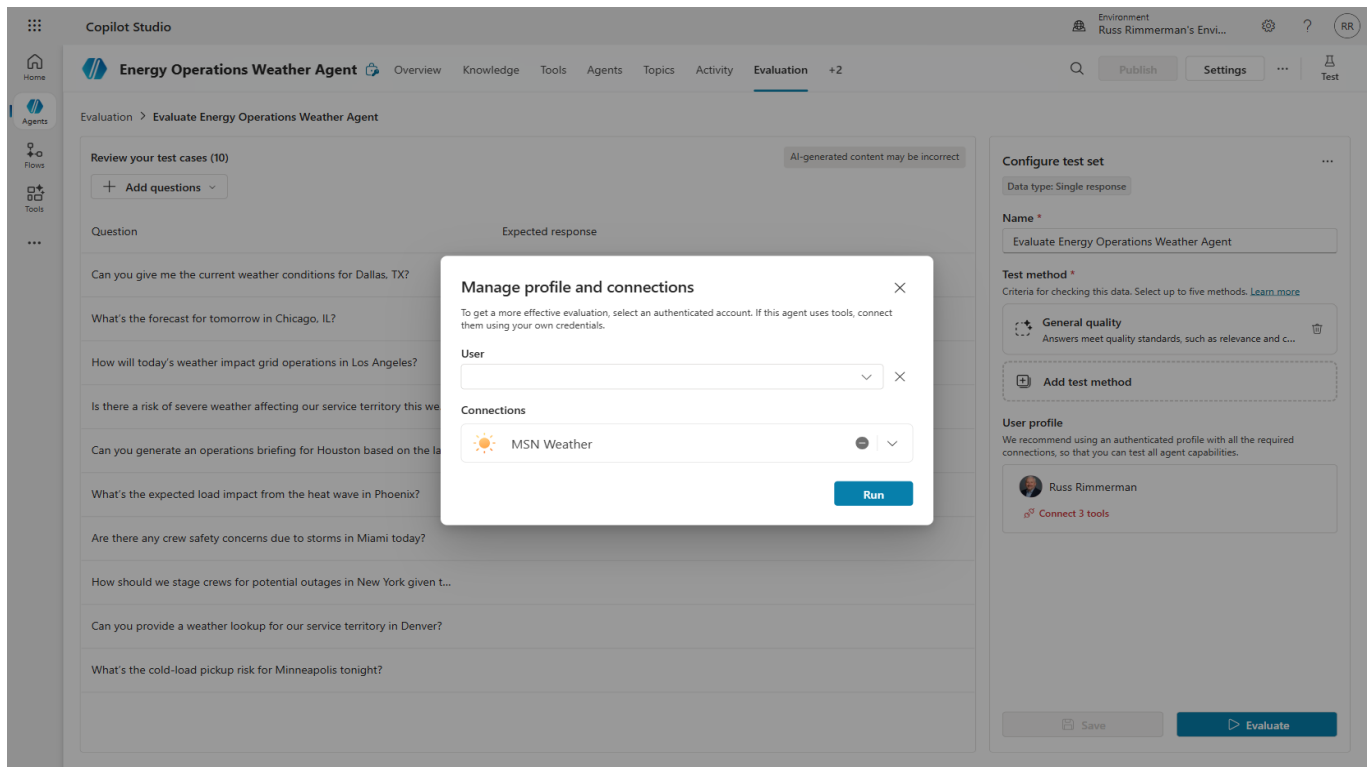
Manage

Save Evaluate

9.2 Configure and Run Evaluation

Step 4: Name the evaluation and select “General quality” as the test method. This checks that responses meet quality standards including relevance and completeness.

Step 5: Click “Evaluate” and select your authenticated account to run the evaluation with real connector access.



Step 6: The evaluation runs each test case through the agent and scores the responses.

The screenshot shows the Copilot Studio interface. The top navigation bar includes 'Home', 'Agents', 'Flows', and 'Tools'. The main header displays 'Energy Operations Weather Agent' with tabs for 'Overview', 'Knowledge', 'Tools', 'Agents', 'Topics', 'Activity', and 'Evaluation'. The 'Evaluation' tab is active, showing a progress bar and a 'Test' button. The main content area is titled 'Evaluation > Evaluate Energy Operations Weather Agent 260617_2326' and contains a large blue circle with the text 'Preparing test cases...'. On the right, the 'Evaluation summary' panel shows the following details:

Evaluation summary	
Final score will appear here once evaluation is completed	
Duration	00:00:08
Test cases	
Test Set	Evaluate Energy Operations Weather Agent
Data type	Single response
User profile	Russ Rimmerman
Tested by	Russ Rimmerman 11:26 PM today

At the bottom of the summary panel is a 'Cancel evaluation' button.

9.3 Review Results

Step 7: Review the evaluation results. The score shows pass/fail rates across all test cases.

Copilot Studio

Environment: Russ Rimmerman's Envi...

Energy Operations Weather Agent

Overview Knowledge Tools Agents Topics Activity Evaluation +2

Evaluation > Evaluate Energy Operations Weather Agent 260617_2326

Test run result

All Pass (4) Fail (6)

Question	Agent response	General quality
Can you give me the current weather conditions for Dallas, TX?	Let's get you connected first, and then I can find t...	Fail
What's the forecast for tomorrow in Chicago, IL?	Let's get you connected first, and then I can find t...	Fail
How will today's weather impact grid operations in Los Angeles?	Let's get you connected first, and then I can find t...	Fail
Is there a risk of severe weather affecting our service territory this week?	To pull up the weather outlook for your service te...	Pass
Can you generate an operations briefing for Houston based on the late...	Let's get you connected first, and then I can find t...	Fail
What's the expected load impact from the heat wave in Phoenix?	Let's get you connected first, and then I can find t...	Pass
Are there any crew safety concerns due to storms in Miami today?	Let's get you connected first, and then I can find t...	Fail
How should we stage crews for potential outages in New York given the...	Let's get you connected first, and then I can find t...	Pass
Can you provide a weather lookup for our service territory in Denver?	I can pull current weather and short-range foreca...	Pass
What's the cold-load pickup risk for Minneapolis tonight?	Let's get you connected first, and then I can find t...	Fail

Evaluation summary

Score: 40%

Pass Fail

AI-generated content may be incorrect

Duration: 00:01:50

Test cases: 10/10

Test Set: Evaluate Energy Operations Weather Agent

Data type: Single response

User profile: Russ Rimmerman

Tested by: Russ Rimmerman 11:26 PM today

NOTE: Connection-related failures (agent asks to connect) are infrastructure issues, not agent quality issues. In production, connections would be pre-established. Focus on the pass cases to validate orchestration logic.

TIP: Re-run evaluations after making changes to measure improvement. Evaluations are your quality gate for production deployment.

Summary and Next Steps

Congratulations! You have built a complete Energy Operations Weather Agent with the following capabilities:

- Real-time weather data retrieval via MSN Weather connectors (3 tools)
- Conversational flow with variables and Power Fx formulas
- AI-powered operations briefing via custom prompt tool
- Multi-agent architecture with Crew Safety Advisor child agent
- Quality assurance via automated evaluation test suites

Key Concepts Learned

- Agent Orchestration: How the AI routes conversations between topics, tools, and child agents
- Power Fx Integration: Using low-code formulas to wire data between components
- Two-Stage AI Pipeline: Retrieval (connector) + Analysis (prompt) pattern
- Connected Agents: Microservices architecture for conversational AI
- Evaluation-Driven Development: Measuring quality before deployment

Suggested Extensions

- Add Knowledge sources (SharePoint documents about grid operations procedures)
- Create additional child agents (e.g., Outage Prediction, Load Forecasting)
- Publish the agent to Microsoft Teams for team-wide access
- Add Adaptive Cards for rich visual responses
- Implement multi-language support for global operations